

AN INTRODUCTION TO NUMBER THEORY



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An Introduction to Number Theory (Veerman)

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CHAPTER OVERVIEW

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1.1: Divisors and Congruences

Definition 1.1

Given two numbers a and b . A multiple b of a is a number that satisfies $b = ac$. A divisor a of b is an integer that satisfies $ac = b$ where c is an integer. We write $a|b$. This reads as a divides b or a is a divisor of b .

Definition 1.2

Let a and b non-zero. The greatest common divisor of two integers a and b is the maximum of the numbers that are divisors of both a and b . It is denoted by $\gcd(a, b)$. The least common multiple of a and b is the least of the positive numbers that are multiples of both a and b . It is denoted by $\text{lcm}(a, b)$.

Note that for any a and b in $\mathbb{Z}$, $\gcd(a, b) \geq 1$, as 1 is a divisor of every integer. Similarly $\text{lcm}(a, b) \leq |ab|$.

Definition 1.3

A number $a > 1$ is prime in $\mathbb{N}$ if its only divisors in $\mathbb{N}$ are a and 1 (the so-called trivial divisors). A number $a > 1$ is composite if it has more than 2 divisors. (The number 1 is neither.)

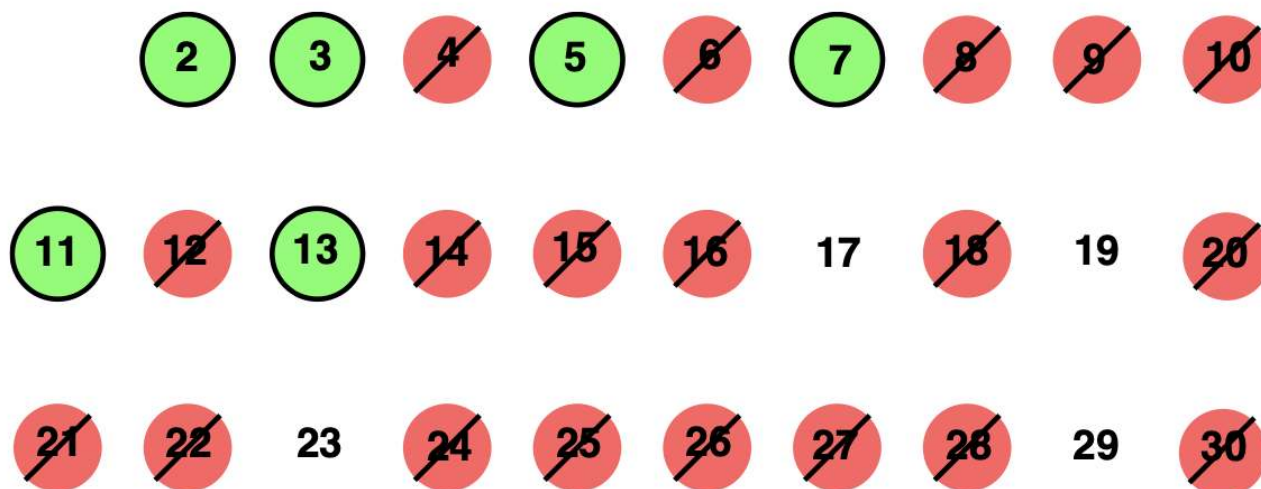


Figure 1. Eratosthenes' sieve up to $n = 30$. All multiples of a less than $\sqrt{31}$ are cancelled. The remainder are the primes less than $n = 31$.

An equivalent definition of prime is a natural number with precisely two (distinct) divisors. *Eratosthenes' sieve* is a simple and ancient method to generate a list of primes for all numbers less than, say, 225. First, list all integers from 2 to 225. Start by circling the number 2 and crossing out all its remaining multiples: 4, 6, 8, etcetera. At each step, circle the smallest unmarked number and cross out all its remaining multiples in the list. It turns out that we need to sieve out only multiples of $\sqrt{225} = 15$ and less (see exercise 2.4). This method is illustrated in Figure 1. When done, the primes are those numbers that are circled or unmarked in the list.

Definition 1.4

Let a and b in $\mathbb{Z}$. Then a and b are relatively prime if $\gcd(a, b) = 1$.

Definition 1.5

Let a and b in $\mathbb{Z}$ and $m \in \mathbb{N}$. Then a is congruent to b modulo m if $a + my = b$ for some $y \in \mathbb{Z}$ or $m|(b - a)$. We write

$$a \equiv_m b \quad \text{or} \quad a \equiv b \pmod{m} \quad \text{or} \quad a \in b + m\mathbb{Z}$$

Definition 1.6

The residue of a modulo m is the (unique) integer r in $\{0, \dots, m-1\}$ such that $a =_m r$. It is denoted by $\text{Res}_m(a)$.

These notions are cornerstones of much of number theory as we will see. But they are also very common in all kinds of applications. For instance, our expressions for the time on the clock are nothing but counting modulo 12 or 24. To figure out how many hours elapse between 4pm and 3am next morning is a simple exercise in working with modular arithmetic, that is: computations involving congruences.

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1.2: Rational and Irrational Numbers

We start with a few results we need in the remainder of this subsection.

Theorem 1.7: Well-ordering Principle

Any non-empty set S in $\mathbb{N} \cup \{0\}$ has a smallest element.

Proof

Suppose this is false. Pick $s_1 \in S$. Then there is another natural number s_2 in S such that $s_2 \leq s_1 - 1$. After a finite number of steps, we pass zero, implying that S has elements less than 0 in it. This is a contradiction.

Note that any non-empty set S of integers with a lower bound can be transformed by addition of a integer $b \in \mathbb{N}_0$ into a non-empty $S + b$ in $\mathbb{N}_0$. Then $S + b$ has a lower bound, and therefore so does S . Furthermore, a non-empty set S of integers with an upper bound can also be transformed into a non-empty $-S + b$ in $\mathbb{N}_0$. Here, $-S$ stands for the collection of elements of S multiplied by -1 . Thus we have the following corollary of the well-ordering principle.

Corollary 1.8

Let be a non-empty set S in $\mathbb{Z}$ with a lower (upper) bound. Then S has a smallest (largest) element.

Definition 1.9

An element $x \in \mathbb{R}$ is called rational if it satisfies $qx - p = 0$ where p and $q \neq 0$ are integers. Otherwise it is called an irrational number. The set of rational numbers is denoted by $\mathbb{Q}$.

The usual way of expressing this, is that a rational number can be written as $\frac{p}{q}$. The advantage of expressing a rational number as the solution of a degree 1 polynomial, however, is that it naturally leads to Definition 1.12.

Theorem 1.10

Any interval in $\mathbb{R}$ contains an element of $\mathbb{Q}$. We say that $\mathbb{Q}$ is dense in $\mathbb{R}$.

Proof

Let $I = (a, b)$ with $b > a$ any interval in $\mathbb{R}$. From Corollary 1.8 we see that there is an n such that $n > \frac{1}{b-a}$. Indeed, if that weren't the case, then $\mathbb{N}$ would be bounded from above, and thus it would have a largest element n_0 . But if $n_0 \in \mathbb{N}$, then so is $n_0 + 1$. This gives a contradiction and so the above inequality must hold.

It follows that $nb - na > 1$. Thus the interval (na, nb) contains an integer, say, p . So we have that $na < p < nb$. The theorem follows upon dividing by n .

The crux of the following proof is that we take an interval and scale it up until we know there is an integer in it, and then scale it back down.

Theorem 1.11

$\sqrt{2}$ is irrational.

Proof

Suppose $\sqrt{2}$ can be expressed as the quotient of integers $\frac{r}{s}$. We may assume that $\gcd(r, s) = 1$ (otherwise just divide out the common factor). After squaring, we get

$$2s^2 = r^2$$

The right hand side is even, therefore the left hand side is even. But the square of an odd number is odd, so r is even. But then r^2 is a multiple of 4. Thus s must be even. This contradicts the assumption that $\gcd(r, s) = 1$.

It is pretty clear who the rational numbers are. But who or where are the others? We just saw that $\sqrt{2}$ is irrational. It is not hard to see that the sum of any rational number plus $\sqrt{2}$ is also irrational. Or that any rational non-zero multiple of $\sqrt{2}$ is irrational. The same holds for $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, etcetera. We look at this in exercise 1.7. From there, is it not hard to see that the irrational numbers are also dense (exercise 1.8). In exercise 1.15, we prove that the number e is irrational. The proof that π is irrational is a little harder and can be found in [1][section 11.17]. In Chapter 2, we will use the fundamental theorem of arithmetic, Theorem 2.14, to construct other irrational numbers. In conclusion, whereas rationality is seen at face value, irrationality of a number may take some effort to prove, even though they are much more numerous as we will see in Section 1.4.

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1.3: Algebraic and Transcendental Numbers

The set of polynomials with coefficients in $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$ is denoted by $\mathbb{Z}[x]$, $\mathbb{Q}[x]$, $\mathbb{R}[x]$ and $\mathbb{C}[x]$, respectively.

Definition 1.12

An element $x \in \mathbb{R}$ is called an algebraic number if it satisfies $p(x) = 0$, where p is a non-zero polynomial in $\mathbb{Z}[x]$. Otherwise it is called a transcendental number.

The transcendental numbers are even harder to pin down than the general irrational numbers. We do know that e and π are transcendental, but the proofs are considerably more difficult (see [12]). We'll see below that the transcendental numbers are far more abundant than the rationals or the algebraic numbers. In spite of this, they are harder to analyze and, in fact, even hard to find. This paradoxical situation where the most prevalent numbers are hardest to find, is actually pretty common in number theory.

The most accessible tool to construct transcendental numbers is Liouville's Theorem. The setting is the following. Given an algebraic number y , it is the root of a polynomial with integer coefficients $f(x) = \sum_{i=0}^d a_i x^i$, where we always assume that the coefficient a_d of the highest power is non-zero. That highest power is called the degree of the polynomial. Note that we can always find a polynomial of higher degree that has y as a root. Namely, multiply f by any other polynomial g .

Definition 1.13

We say that $f(x) = \sum_{i=0}^d a_i x^i$ in $\mathbb{Z}[x]$ is a minimal polynomial for ρ if f is a non-zero polynomial of minimal degree such that $f(\rho) = 0$.

Theorem 1.14 Liouville's Theorem

Let f be a minimal polynomial of degree $d \geq 2$ for $\rho \in \mathbb{R}$. Then

$$\exists c(\rho) > 0 \text{ such that } \forall \frac{p}{q} \in \mathbb{Q} : \left| \rho - \frac{p}{q} \right| > \frac{c(\rho)}{q^d}$$

Proof

Clearly, if $\left| \rho - \frac{p}{q} \right| \geq 1$, the inequality is satisfied. So assume that $\left| \rho - \frac{p}{q} \right| < 1$.

Now let f be a minimal polynomial for ρ , and set

$$K = \max_{t \in [\rho-1, \rho+1]} |f'(t)|$$

We know that $f(\frac{p}{q})$ is not zero, because otherwise f would have a factor $(x - \frac{p}{q})$. In that case, the quotient g of f and $(x - \frac{p}{q})$ would not necessarily have integer coefficients, but some integral multiple mg of g would. However, mg would be of lower degree, thus contradicting the minimality of f . This gives us that

$$|q^d f(\frac{p}{q})| = \left| \sum_{i=0}^d a_i p^i q^{d-i} \right| \geq 1 \Rightarrow |f(\frac{p}{q})| \geq q^{-d}$$

because it is a non-zero integer. Finally, we use the mean value theorem which tells us that there is a t between ρ and $\frac{p}{q}$ such that

$$K \geq |f'(t)| = \left| \frac{f(\frac{p}{q}) - f(\rho)}{\frac{p}{q} - \rho} \right| \geq \frac{q^{-d}}{\left| \frac{p}{q} - \rho \right|}$$

since $f(\rho) = 0$. For K as above, this gives us the desired inequality.

Definition 1.15

A real number ρ is called a Liouville number if for all $n \in \mathbb{N}$, there is a rational number $\frac{p}{q}$ such that

$$\left| \rho - \frac{p}{q} \right| < \frac{1}{q^n}$$

It follows directly from Liouville's theorem that such numbers must be transcendental. Liouville numbers can be constructed fairly easily. The number

$$\rho = \sum_{k=1}^{\infty} 10^{-k!}$$

is an example. If we set $\frac{p}{q}$ equal to $\sum_{k=1}^n 10^{-k!}$, then $q = 10^{n!}$. Then

$$\left| \rho - \frac{p}{q} \right| = \sum_{k=n+1}^{\infty} 10^{-k!}$$

It is easy to show that this is less than q^{-n} (exercise 1.17).

It is worth noting that there is an optimal version of Liouville's Theorem. We record it here without proof.

Theorem 1.16 Roth's Theorem

Let $\rho \in \mathbb{R}$ be algebraic. Then for all $\epsilon > 0$

$$\exists c(\rho, \epsilon) > 0 \text{ such that } \forall \frac{p}{q} \in \mathbb{Q} : \left| \rho - \frac{p}{q} \right| > \frac{c(\rho, \epsilon)}{q^{2+\epsilon}}$$

where $c(\rho, \epsilon)$ depends only on ρ and ϵ .

This result is all the more remarkable if we consider it in the context of the following more general result (which we will have occasion to prove in Chapter 6).

Theorem 1.17

Let $\rho \in \mathbb{R}$ be irrational. Then there are infinitely many $\frac{p}{q} \in \mathbb{Q}$ such that $\left| \rho - \frac{p}{q} \right| < \frac{1}{q^2}$.

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1.4: Countable and Uncountable Sets

Definition 1.18

A set S is countable if there is a bijection $f : \mathbb{N} \rightarrow S$. An infinite set for which there is no such bijection is called uncountable.

Proposition 1.19

Every infinite set S contains a countable subset.

Proposition 1.19

Every infinite set S contains a countable subset.

Proof

Choose an element s_1 from S . Now $S - \{s_1\}$ is not empty because S is not finite. So, choose s_2 from $S - \{s_1\}$. Then $S - \{s_1, s_2\}$ is not empty because S is not finite. In this way, we can remove s_{n+1} from $S - \{s_1, s_2, \dots, s_n\}$ for all n . Then set $\{s_1, s_2, \dots\}$ is countable and is contained in S .

So countable sets are the smallest infinite sets in the sense that there are no infinite sets that contain no countable set. But there certainly are larger sets, as we will see next.

Theorem 1.20

The set $\mathbb{R}$ is uncountable.

Proof

The proof is one of mathematics' most famous arguments: Cantor's diagonal argument [8]. The argument is developed in two steps.

Let T be the set of semi-infinite sequences formed by the digits 0 and 2. An element $t \in T$ has the form $t = t_1 t_2 t_3 \dots$ where $t_i \in \{0, 2\}$. The first step of the proof is to prove that T is uncountable. So suppose it is countable. Then a bijection t between $\mathbb{N}$ and T allows us to uniquely define the sequence $t(n)$, the unique sequence associated to n . Furthermore, they form an exhaustive list of the elements of T . For example,

$$t(1) = 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, \dots$$

$$t(2) = 2, 0, 2, 0, 2, 0, 2, 0, 2, 2, \dots$$

$$t(3) = 0, 0, 0, 2, 2, 2, 2, 2, 2, 2, \dots$$

$$t(4) = 2, 2, 2, 2, 2, 2, 0, 0, 0, 0, \dots$$

$$t(5) = 0, 0, 0, 2, 2, 0, 2, 0, 0, 2, 0, \dots$$

$$t(6) = 2, 0, 0, 0, 0, 2, 0, 0, 0, 2, 2, \dots$$

$\vdots$

Construct t^* as follows: for every n , its n th digit differs from the n th digit of $t(n)$. In the above example, $t^* = 2, 2, 2, 0, 2, 0, \dots$. But now we have a contradiction, because the element t^* cannot occur in the list. In other words, there is no surjection from $\mathbb{N}$ to T . Hence there is no bijection between $\mathbb{N}$ and T .

The second step is to show that there is a subset K of $\mathbb{R}$ such that there is no surjection (and thus no bijection) from $\mathbb{N}$ to K . Let t be a sequence with digits t_i . Define $f : T \rightarrow \mathbb{R}$ as follows

$$f(t) = \sum_{i=1}^{\infty} t_i 3^{-i}$$

If s and t are two distinct sequences in T , then for some k they share the first $k - 1$ digits but $t_k = 2$ and $s_k = 0$. So

$$f(t) - f(s) = 2 \cdot 3^{-k} + \sum_{i=k+1}^{\infty} (t_i - s_i) 3^{-i} \geq 2 \cdot 3^{-k} - 2 \sum_{i=k+1}^{\infty} 3^{-i} = 3^{-k}$$

Thus f is injective. Therefore f is a bijection between T and the subset $K = f(T)$ of $\mathbb{R}$. If there is a surjection g from $\mathbb{N}$ to $K = f(T)$, then,

$$\mathbb{N} \xrightarrow{g} K \xleftarrow{f} T$$

And so $f^{-1}g$ is a surjection from $\mathbb{N}$ to T . By the first step, this is impossible. Therefore, there is no surjection g from $\mathbb{N}$ to K , much less from $\mathbb{N}$ to $\mathbb{R}$.

Corollary 1.21

- (i) The set of infinite sequences in $\{1, 2, \dots, b - 1\}^{\mathbb{N}}$ is uncountable.
- (ii) The set of finite sequences (but without bound) in $\{1, 2, \dots, b - 1\}^{\mathbb{N}}$ is countable.

Proof

The proof of (i) is the same as the proof that T is uncountable in the proof of Theorem 1.20. The proof of (ii) consists of writing first all b words of length 1, then all b^2 words of length 2, and so forth. Every finite string will occur in the list.

Theorem 1.22

- (i) The set $\mathbb{Z}^2$ is countable.
- (ii) $\mathbb{Q}$ is countable.

Proof

- (i) The proof relies on Figure 2. In it, a directed path γ is traced out that passes through all points of $\mathbb{Z}^2$. Imagine that you start at $(0, 0)$ and travel along γ with unit speed. Keep a counter $c \in \mathbb{N}$ that marks the point $(0, 0)$ with a "1". Up the value of the counter by 1 whenever you hit a point of $\mathbb{Z}^2$. This establishes a bijection between $\mathbb{N}$ and $\mathbb{Z}^2$.

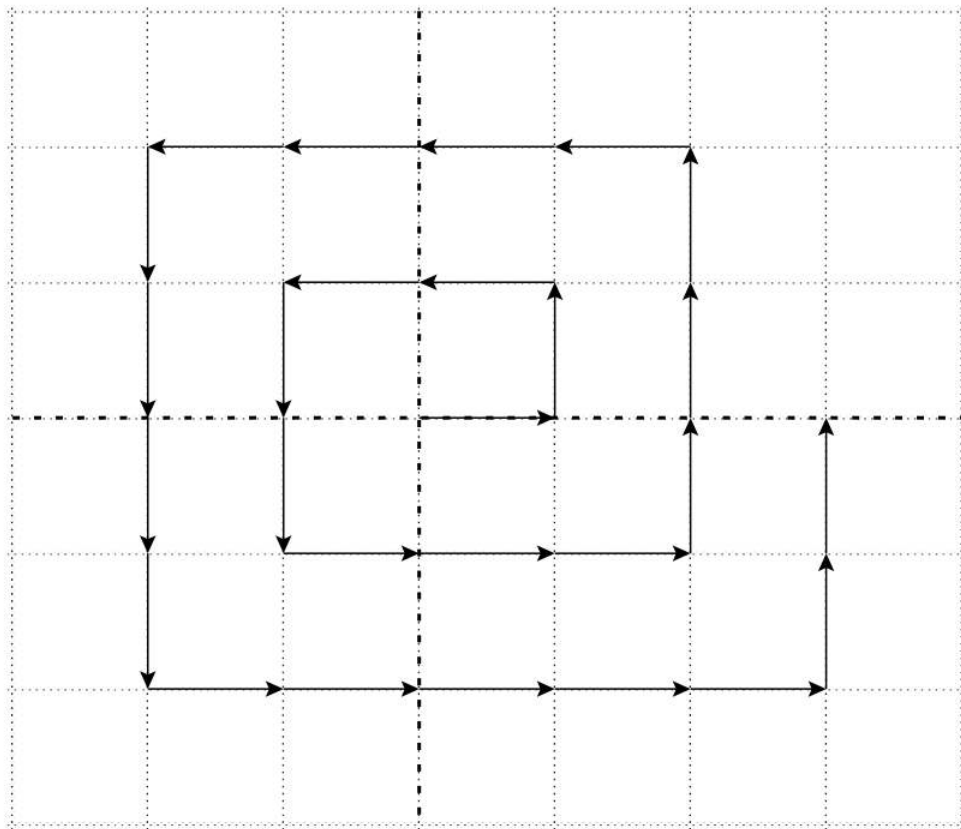


Figure 2. A directed path γ passing through all points of $\mathbb{Z}^2$.

(ii) Again travel along γ with unit speed. Keep a counter $c \in \mathbb{N}$ that marks the point $(0, 1)$ with a “1”. Up the value of the counter by 1. Continue to travel along the path until you hit the next point (p, q) that is not a multiple of any previous and such q is not zero. Mark that point with the value of the counter. $\mathbb{Q}$ contains $\mathbb{N}$ and so is infinite. Identifying each marked point (p, q) with the rational number $\frac{p}{q}$ establishes the countability of $\mathbb{Q}$.

Notice that this argument really tells us that the product of a countable set and another countable set is still countable. The same holds for any finite product of countable set. Since an uncountable set is strictly larger than a countable, intuitively this means that an uncountable set must be a lot larger than a countable set. In fact, an extension of the above argument shows that the set of algebraic numbers is countable. And thus, in a sense, it forms a small subset of all reals. All the more remarkable, that almost all reals that we know anything about are algebraic numbers, a situation we referred to at the end of Section 1.4.

It is useful and important to have a more general definition of when two sets “have the same number of elements”.

Definition 1.23

Two sets A and B are said to have the same cardinality if there is a bijection $f : A \rightarrow B$. It is written as $|A| = |B|$. If there is an injection $f : A \rightarrow B$, then $|A| \leq |B|$.

Definition 1.24

An equivalence relation on a set A is a (sub)set $\mathbb{R}$ of ordered pairs in $A \times A$ that satisfy three requirements.

- $(a, a) \in \mathbb{R}$ (reflexivity).
- If $(a, b) \in \mathbb{R}$, then $(b, a) \in \mathbb{R}$ (symmetry).
- If $(a, b) \in \mathbb{R}$ and $(b, c) \in \mathbb{R}$, then $(a, c) \in \mathbb{R}$ (transitivity).

Usually $(a, b) \in \mathbb{R}$ is abbreviated to $a \sim b$. The mathematical symbol “=” is an equivalence.

It is easy to show that having the same cardinality is an equivalence relation on sets (exercise 1.23). Note that the cardinality of a finite set is just the number of elements it contains. An excellent introduction to the cardinality of infinite sets in the context of naive set theory can be found in [15].

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1.5: Exercises

Exercise 1.5.1

Apply Eratosthenes' Sieve to get all prime numbers between 1 and 200. (Hint: you should get 25 primes less than 100, and 21 between 100 and 200.)

Exercise 1.5.2

Factor the following into prime numbers (write as a product of primes): 393, 16000, 5041, 1111, 1763, 720.

Exercise 1.5.3

Find pairs of primes that differ by 2. These are called twin primes. Are there infinitely many such pairs? (Hint: This is an open problem; the affirmative answer is called the twin prime conjecture.)

Exercise 1.5.4: Goldbach Conjecture

Show that small enough even integers can be written as the sum of two primes. Is this always true? (Hint: This is an open problem; the affirmative answer is called the Goldbach conjecture.)

Exercise 1.5.5

Comment on the types of numbers (rational, irrational, transcendental) we use in daily life.

- What numbers do we use to pay our bills?
- What numbers do we use in computer simulations of complex processes?
- What numbers do we use to measure physical things?
- Give examples of the usage of the "other" numbers.

Exercise 1.5.6

Let a and b be rationals and x and y irrationals.

- Show that ax is irrational iff $a \neq 0$.
- Show that $b + x$ is irrational.
- Show that $ax + b$ is irrational iff $a \neq 0$.
- Conclude that $a\sqrt{2} + b$ is irrational iff $a \neq 0$.

Exercise 1.5.7

Show that $\sqrt{3}$, $\sqrt{5}$, etcetera (square roots of primes) are irrational.

Exercise 1.5.8

Show that numbers of the form that $a\sqrt{2} + b$ are irrational and dense in the reals (a and b are rational).

Exercise 1.5.9

- Use an pictorial argument similar to that of Figure 2 to show that $\mathbb{N} \times \mathbb{N}$ (the set of lattice points (n, m) with n and m in $\mathbb{N}$) is countable.
- Suppose A_i are countably infinite sets where $i \in I$ and I countable. Show that there is a bijection $f_i : \mathbb{N} \rightarrow A_i$ for each i .
- Show there is a bijection $F : \mathbb{N} \times \mathbb{N} \rightarrow \bigcup_{i \in I} A_i$ given by $F(n, m) = f_n(m)$.
- Conclude that the countable union of countably infinite sets is countably infinite.

Exercise 1.5.10

What is wrong in the following attempt to prove that $[0, 1]$ is uncountable?

Assume that $[0, 1]$ is countable, that is: there is a bijection f between $[0, 1]$ and $\mathbb{N}$. Let $r(n)$ be the unique number in $[0, 1]$ assigned to n . Thus the infinite array $(r(1), r(2), \dots)$ forms an exhaustive list of the numbers in $[0, 1]$, as follows:

$$r(1) = 0.000000000000 \dots$$

$$r(2) = 0.10101010111 \dots$$

$$r(3) = 0.00011111111 \dots$$

$$r(4) = 0.11111100000 \dots$$

$$r(5) = 0.00010010010 \dots$$

$$r(6) = 0.10000100011 \dots$$

$\vdots$

(Written as number on the base 2.) Construct r^* as the string whose n th digit differs from that of $r(n)$. Thus in this example: $r^* = 0.111010 \dots$,

which is different from all the other listed binary numbers in $[0, 1]$. (Hint: what if r^* ends with an infinite all ones subsequence?)

Exercise 1.5.11

The set $f(T)$ in the proof of Theorem 1.20 is called the middle third Cantor set. Find its construction. What does it look like? (Hint: locate the set of numbers whose first digit (base 3) is a 1; then the set of numbers whose second digit is a 1.)

Exercise 1.5.12

The integers exhibit many, many other intriguing patterns. Given the following function:

$$\begin{cases} n \text{ even:} & f(n) = \frac{n}{2} \\ n \text{ odd:} & f(n) = \frac{3n+1}{2} \end{cases} \quad (1.5.1)$$

- (Periodic orbit) Show that f sends 1 to 2 and 2 to 1.
- (Periodic orbit attracts) Show that if you start with a small positive integer and apply f repeatedly, eventually you fall on the orbit in (a).
- Show that this is true for all positive integers.
(Hint: This is an open problem; the affirmative answer is called the Collatz conjecture.)

Exercise 1.5.13

It is known that $2^{11213} - 1$ is prime. How many decimal digits does this number have? (Hint: $\log_{10} 2 \approx 0.301029996$)

Exercise 1.5.14

This exercise prepares for Mersenne and Fermat primes, see Definition 5.11.

- Use $\sum_{i=0}^{b-1} 2^{ia} = \frac{2^{ab}-1}{2^a-1}$ to show that if $2^p - 1$ is prime, then p must be prime.
- Use $\sum_{i=0}^{b-1} (-2)^{ia} = \frac{(-2)^{ab}-1}{(-2)^a-1}$ to show that if $2^p + 1$ is prime, then p has no odd factor.

Exercise 1.5.15

In the following, we assume that $e - 1 = \sum_{i=1}^{\infty} \frac{1}{i!} = \frac{p}{q}$ is rational, and show that this leads to a contradiction.

a. Show that the above assumption implies that

$$\sum_{i=1}^q \frac{q!}{i!} + \frac{q!}{q+i!} = p(q-1)!$$

(Hint: multiply both sides of by $q!$.)

b. Show that $\sum_{i=1}^q \frac{q!}{i!} + \frac{q!}{q+i!} < \frac{1}{q^2}$. (Hint: write out a few terms of the sum on the left.)

c. Show that the sums in (b) cannot have an integer value.

d. Show that the other two terms in (a) have an integer value.

e. Conclude there is a contradiction unless the assumption that e is rational is false.

Exercise 1.5.16

Show that Liouville's theorem (Theorem 1.14) also holds for rational for rational numbers $\rho = \frac{r}{s}$ as long as $\frac{p}{q} \neq \frac{r}{s}$.

Exercise 1.5.17

a. Show that for all positive integers p and n , we have $p(n+1)n! \leq (n+p)!$.

b. Use (a) to show that

$$\sum_{k=n+1}^{\infty} 10^{-k!} \leq \sum_{k=n+1}^{\infty} 10^{-p(n+1)n!} (1 - 10^{-p(n+1)n!})^{-1}$$

c. Show that b) implies equation (1.1).

Exercise 1.5.18

Show that the inequality of Roth's theorem does not hold for all numbers. (Hint: Let ρ be a Liouville number.)

Definition 1.25

Let A be a set. Its power set $P(A)$ is the set whose elements are the subsets of A . This always includes the empty set denoted by $\emptyset$.

In the next two exercises, the aim is to show something that is obvious for finite sets, namely:

Theorem 1.26

The cardinality of a power set is always (strictly) greater than that of the set itself.

Exercise 1.5.19

a) Given a set A , show that there is an injection $f : A \rightarrow P(A)$. (Hint: for every element $a \in A$ there is a set $\{a\}$.)

b) Conclude that $|A| \leq |P(A)|$. (Hint: see Definition 1.23.)

Exercise 1.5.20

Let A be an arbitrary set. Assume that there is a surjection $S : A \rightarrow P(A)$ and define

$$R = \{a \in A \mid a \notin S(a)\}$$

a. Show that there is a $q \in A$ such that $S(q) = R$.

b. Show that if $q \in R$, then $q \notin R$. (Hint: equation 1.2.)

- c. Show that if $q \notin R$, then $q \in R$. (Hint: equation 1.2.)
- d. Use (b) and (c) and exercise 1.19, to establish that $|A| < |P(A)|$. (Hint: see Definition 1.23.)

In the next two exercises we show that the cardinality of $\mathbb{R}$ equals that of $P(\mathbb{N})$. This implies that that $|\mathbb{R}| > |\mathbb{N}|$, which also follows from Theorem 1.20.

Exercise 1.5.21

Let T be the set of sequences defined in the proof of Theorem 1.20. To a sequence $t \in T$, associate a set $S(t)$ in $P(\mathbb{N})$ as follows:

$$i \in S \text{ if } t(i) = 2 \quad \text{and} \quad i \notin S \text{ if } t(i) = 0$$

- a. Show that there is a bijection $S : T \rightarrow P(\mathbb{N})$.
- b. Use the bijection f in the proof of Theorem 1.20 to show there is a bijection $K \rightarrow P(\mathbb{N})$.
- c. Show that (a) and (b) imply that $|P(\mathbb{N})| = |K| = |T|$. (Hint: see Definition 1.23.)
- d. Find an injection $K \rightarrow \mathbb{R}$ and conclude that $|P(\mathbb{N})| \leq |\mathbb{R}|$.

Exercise 1.5.22

- a. Show that there is a bijection $\mathbb{R} \rightarrow (0, 1)$.
- b. Show that there is an injection $(0, 1) \rightarrow T$. (Hint: use usual binary (base 2) expansion of reals.)
- c. Use (a), (b), and exercise 1.21 (a), to show that $|\mathbb{R}| \leq |P(\mathbb{N})|$.
- d. Use (c) and exercise 1.21 (d) to show that $|\mathbb{R}| = |P(\mathbb{N})|$.

Exercise 1.5.23

- a. Show that having the same cardinality (see Definition 1.23) is an equivalence relation on sets.
- b. Conclude that “cardinality” is an “equivalence class of sets”.

Exercise 1.5.24

- a. Fix some $n > 0$. Show that having the same remainder modulo n is an equivalence relation on $\mathbb{Z}$. (Hint: for example, $-8, 4$, and 16 have the remainder modulo 12 .)
- b. Show that addition respects this equivalence relation. (Hint: If $a + b = c$, $a \sim a'$, and $b \sim b'$, then $a' + b' = c'$ with $c \sim c'$.)
- c. The same question for multiplication.

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CHAPTER OVERVIEW

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2.1: Bézout's Lemma

Definition 2.1

The floor of a real number θ is defined as follows: $\lfloor \theta \rfloor$ is the greatest integer less than or equal to θ . The fractional part $\{\theta\}$ of the number θ is defined as $\theta - \lfloor \theta \rfloor$. Similarly, the ceiling of θ , $\lceil \theta \rceil$, gives the smallest integer greater than or equal to θ .

By the well-ordering principle, Corollary 1.8, the number $\lfloor \theta \rfloor$ and $\lceil \theta \rceil$ exist for any $\theta \in \mathbb{R}$.

Definition 2.2

Given a number $\xi \in \mathbb{R}$, we denote its norm (or absolute value) $|\xi|$ by $N(\xi)$.

This last definition seems clumsy and unnecessary in the present context. But it will save us much trouble later on (see Section ??).

Lemma 2.3

Given r_1 and r_2 with $r_2 > 0$, then there are unique q_2 and $r_3 \geq 0$ with $N(r_3) < N(r_2)$ such that $r_1 = r_2 q_2 + r_3$.

Proof

Noting that $\frac{r_1}{r_2}$ is a rational number, we can choose the integer $q_2 = \lfloor \frac{r_1}{r_2} \rfloor$ so that

$$\frac{r_1}{r_2} = q_2 + \epsilon$$

where $\epsilon \in [0, 1)$. The integer q_2 is called the quotient. Multiplying by r_2 gives the result.

Note that $r_3 \in \{0, \dots, r_2 - 1\}$. Thus among other things, this lemma implies that every integer has a unique residue (see Definition 1.6). If $N(r_1) < N(r_2)$, then $q_2 = 0$. In this case, $\epsilon = \frac{r_1}{r_2}$ and we learn nothing new. But if $N(r_1) > N(r_2)$, then $q_2 \neq 0$ and we have written r_1 as a multiple of r_2 plus a remainder r_3 .

Definition 2.4

Given r_1 and r_2 with $r_2 > 0$, the computation of q_2 and r_3 satisfying Lemma 2.3 is called the division algorithm. Note that $r_3 = \text{Res}_{r_2}(r_1)$.

Theorem 2.5 (Bezout's Lemma)

Let a and b be such that $\gcd(a, b) = d$. Then $ax + by = c$ has integer solutions for x and y if and only if c is a multiple of d .

Proof

Let S and $\nu(S)$ be the sets:

$$S = \{ax + by \mid x, y \in \mathbb{Z}, ax + by \neq 0\}$$

.

$$\nu(S) = \{N(s) \mid s \in S\} \subseteq \mathbb{N}$$

Then $\nu(S) \neq 0$ and is bounded from below. Thus by the well-ordering principle of $\mathbb{N}$, it has a smallest element e . Then there is an element $d \in S$ that has that norm: $N(d) = e$.

For that d , we use the division algorithm to establish that there are q and $r \geq 0$ such that

$$a = qd + r \quad \text{and} \quad N(r) < N(d)$$

Now substitute $d = ax + by$. A short computation shows that r can be rewritten as:

$$r = a(1 - qx) + b(qy)$$

This shows that $r \in S$. But we also know from (2.1) that $N(r)$ is smaller than $N(d)$. Unless $r = 0$, this is a contradiction because of the way d is defined. But $r = 0$ implies that d is a divisor of a . The same argument shows that d is also a divisor of b . Thus d is a common divisor of both a and b .

Now let e be any divisor of both a and b . Then $e|(ax + by)$, and so $e|d$. But if $e|d$, then $N(e)$ must be smaller than or equal to $N(d)$. Therefore, d is the greatest common divisor of both a and b .

By multiplying x and y by f , we achieve that for any multiple fd of d that

$$afx + bfy = fd$$

On the other hand, let d be as defined above and suppose that x, y , and c are such that

$$ax + by = c$$

Since d divides a and b , we must have that $d|c$, and thus c must be a multiple of d .

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2.2: Corollaries of Be'zout's Lemma

Lemma 2.6 (Euclid's Lemma)

Let a and b be such that $\gcd(a, b) = 1$ and $a|bc$. Then $a|c$.

Proof

By Be'zout, there are x and y such that $ax + by = 1$. Multiply by c to get:

$$acx + bcy = c$$

Since $a|bc$, the left hand side is divisible by a , and so is the right hand side.

Euclid's lemma is so often used, that it will pay off to have a few of the standard consequences for future reference.

Theorem 2.7 Cancellation Theorem

Let $\gcd(a, b) = 1$ and b positive. Then $ax =_b ay$ if and only if $x =_b y$.

Proof

The statement is trivially true if $b = 1$, because all integers are equal modulo 1.

If $ax =_b ay$, then $a(x - y) =_b 0$. The latter is equivalent to $b|a(x - y)$. The conclusion follows from Euclid's Theorem. Vice versa, if $x =_b y$, then $(x - y)$ is a multiple of b and $a(x - y)$ is a multiple of b .

Used as we are to cancellations in calculations in $\mathbb{R}$, it is easy to underestimate the importance of this result. As an example, consider solving $21x =_{35} 21y$. It is tempting to say that this implies that $x =_{35} y$. But in fact, $\gcd(21, 35) = 7$ and the solution set is $x =_5 y$, as is easily checked. This example is in fact a special of the following corollary.

Corollary 2.8

Let $\gcd(a, b) = d$ and b positive. Then $ax =_b ay$ if and only if $x =_{b/d} y$.

Proof

Again the statement is equivalent to $b|a(x - y)$. But now we can divide by d to get $\frac{b}{d} | \frac{a}{d}(x - y)$. Now we can apply the cancellation theorem.

Corollary 2.9

Let $\gcd(a, b) = 1$ and b positive. If $a|c$ and $b|c$ then $ab|c$.

Proof

We have $c = ax = by$ and thus $ax =_b 0$. The cancellation theorem implies that $x =_b 0$.

Proposition 2.10

$ax =_m c$ has a solution if and only if $\gcd(a, m)|c$.

Proof

By Definition 1.5, $ax =_m c$ means $ax + my = c$ for some y , and the result follows from Bezout.

Corollary 2.11

For any $n \geq 1$, if p is prime and $p \mid \prod_{i=1}^n a_i$, then there is $j \leq n$ such that $p \mid a_j$.

Proof

We prove this by induction on n , the number of terms in the product. Let $S(n)$ be the statement of the Corollary.

The statement $S(1)$ is: If p is prime and $p \mid a_1$, then $p \mid a_1$, which is trivially true.

For the induction step, suppose that for any $k > 1$, $S(k)$ is valid and let $p \mid \prod_{i=1}^{k+1} a_i$. Then,

$$p \mid \left(\left(\prod_{i=1}^k a_i \right) a_{k+1} \right)$$

Applying Euclid's Lemma, it follows that,

$$p \mid \prod_{i=1}^k a_i \quad \text{or, if not, then} \quad p \mid a_{k+1}$$

In the former case $S(k+1)$ holds because $S(k)$ does. In the latter, we see that $S(k+1)$ also holds.

Corollary 2.12

If p and q_i are prime and $p \mid \prod_{i=1}^n q_i$, then there is $j \leq n$ such that $p = q_j$.

Proof

Corollary 2.11 says that if p and all q_i are primes, then there is $j \leq n$ such that $p \mid q_j$. Since q_j is prime, its only divisor are 1 and itself. Since $p \neq 1$ (by the definition of prime), $p = q_j$.

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2.3: The Fundamental Theorem of Arithmetic

The last corollary of the previous section enables us to prove the most important result of this chapter. But first, we introduce units and extend the definition of primes to $\mathbb{Z}$.

Definition 2.13

Units in $\mathbb{Z}$ are -1 and 1 . All other numbers are non-units. A number $n \neq 0$ in $\mathbb{Z}$ is called composite if it can be written as a product of two non-units. If n is not 0 , not a unit and not composite, it is a prime.

Theorem 2.14 (The Fundamental Theorem of Arithmetic)

Every nonzero integer $n \in \mathbb{Z}$

1. is a product of powers of primes (up to a unit) and
2. that product is unique (up to the order of multiplication and up to multiplication by the units ± 1).

Remark: The theorem is also called the unique factorization theorem. Its statement means that up to re-ordering of the p_i , every integer n can be uniquely expressed as

$$n = \pm 1 \prod_{i=1}^r p_i^{l_i}$$

where the p_i are distinct primes.

Proof

Statement (1): Define S to be the set of integers n that are not products of primes times a unit, and the set $\nu(S)$ their norms. If the set S is non-empty, then then by the well-ordering principle (Theorem 1.7), $\nu(S)$ has a smallest element. Let a be one of the elements in S that minimize $\nu(S)$. By possibly multiplying by -1 , we may assume that a is positive.

If a is prime, then it can be factored into primes, namely $a = a$, and the theorem would hold in this case. Thus a is a composite number, $a = bc$ and both b and c are non-units. Thus $N(b)$ and $N(c)$ are strictly smaller than $N(a)$. If b and c are products of primes, then, of course, so is $a = bc$. Therefore at least one of a or b is not. But this contradicts the assumptions on a .

Statement (2): Let S be the set of integers that have more than one factorization and $\nu(S)$ the set of their norms. If the set S is non-empty, then by the well-ordering principle (Theorem 1.7), $\nu(S)$ has a smallest element. Let a be one of the elements in S that minimize $\nu(S)$.

Thus we have

$$a = u \prod_{i=1}^r p_i = u' \prod_{i=1}^r p'_i$$

where at least some of the p_i and p'_i do not match up. Here, u and u' are units. Clearly, p_1 divides a . By Corollary 2.12, p_1 equals one of the p'_i , say, p'_1 . Since primes are not units, $N(\frac{a}{p_1})$ is strictly less than $N(a)$. Therefore, by hypothesis, $\frac{a}{p_1}$ is uniquely factorizable. But then then, the primes in

$$\frac{a}{p_1} = u \prod_{i=2}^r p_i = u' \prod_{i=2}^r p'_i$$

all match up (up to units).

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2.4: Corollaries of the Fundamental Theorem of Arithmetic

The unique factorization theorem is intuitive and easy to use. It is very effective in proving a great number of results. Some of these results can be proved with a little more effort without using the theorem (see exercise 2.5 for an example).

Corollary 2.15

For all a and b in $\mathbb{Z}$ not both equal to 0, we have that $\gcd(a, b) \cdot \text{lcm}(a, b) = ab$ up to units.

Proof

If c is a divisor or multiple of a , then so is $-c$. So a and $-a$ have the same divisors and multiples. Therefore without loss of generality we may assume that a and b are positive.

Given two positive numbers a and b , let $P = \{p_i\}_{i=1}^k$ be the list of all prime numbers occurring in the unique factorization of a or b . We then have:

$$a = \prod_{i=1}^s p_i^{k_i} \quad \text{and} \quad b = \prod_{i=1}^s p_i^{l_i}$$

where k_i and l_i in $\mathbb{N} \cup \{0\}$. Now define:

$$m_i = \min(k_i, l_i) \quad \text{and} \quad M_i = \max(k_i, l_i)$$

and let the numbers m and M be given by

$$m = \prod_{i=1}^s p_i^{m_i} \quad \text{and} \quad M = \prod_{i=1}^s p_i^{M_i}$$

Since $m_i + M_i = k_i + l_i$, it is clear that the multiplication $m \cdot M$ yields ab .

Now all we need to do, is showing that m equals $\gcd(a, b)$ and that M equals $\text{lcm}(a, b)$. Clearly m divides both a and b . On the other hand, any integer greater than m has a unique factorization that either contains a prime not in the list P and therefore divides neither a nor b , or, if not, at least one of the primes in P in its factorization has a power greater than m_i . In the last case m is not a divisor of at least one of a and b . The proof that M equals $\text{lcm}(a, b)$ is similar.

A final question one might ask, is how many primes are there? In other words, how long can the list of primes in a factorization be? Euclid provided the answer around 300BC.

Theorem 2.16 (Infinity of Primes)

There are infinitely many primes.

Proof

Suppose the list P of all primes is finite, so that $P = \{p_i\}_{i=1}^n$. Define the integer d as the product of all primes (to the power 1):

$$d = \prod_{i=1}^n p_i$$

If $d + 1$ is a prime, we have a contradiction. So $d + 1$ must be divisible by a prime p_i in P . But then we have

$$p_i | d \quad \text{and} \quad p_i + 1 | d$$

But since $(d + 1)(1) + d(-1) = 1$, Bezout's lemma implies that $\gcd(d, d + 1) = 1$, which contradicts our earlier conclusion $p_i | d$.

One the best known consequences of the fundamental theorem of arithmetic is probably the theorem that follow below. A special case, namely $\sqrt{2}$ is irrational (see theorem 1.11), was known to Pythagoras in the 6th century BC.

Theorem 2.17

Let $n > 0$ and $k > 1$ be integers. Then $n^{\frac{1}{k}}$ is either an integer or irrational.

Proof

Assume $n^{\frac{1}{k}}$ is rational. That is: suppose that there are integers a and b such that

$$n^{\frac{1}{k}} = \frac{a}{b} \Rightarrow n \cdot b^k = a^k$$

Divide out any common divisors of a and b , so that $\gcd(a, b) = 1$. Then by the fundamental theorem of arithmetic:

$$n \prod_{i=1}^s p_i^{k m_i} = \prod_{i=s+1}^r p_i^{k l_i}$$

Therefore, in the factorization of n , each prime p_i must occur with a power that is a non-negative multiple of k . Because $\gcd(a, b) = 1$, the prime p_i on the left and right side are distinct. This is only possible if $\prod_{i=1}^s p_i^{k m_i}$ equals 1. But then n is the k -th power of an integer.

Lemma 2.18

We have

$$\forall i \in \{1, \dots, n\} : \gcd(a_i, b) = 1 \Leftrightarrow \gcd\left(\prod_{i=1}^n a_i, b\right) = 1$$

Proof

The easiest way to see this uses prime power factorization. If $\gcd(\prod_{i=1}^n a_i, b) = d > 1$, then d contains a factor $p > 1$ that is a prime. Since p divides $\prod_{i=1}^n a_i$, at least one of the a_i must contain (by Corollary 2.11) a factor p . Since p also divides b , this contradicts the assumption that $\gcd(a_i, b) = 1$.

Vice versa, if $\gcd(a_i, b) = d > 1$ for some i , then also $\prod_{i=1}^n a_i$ is divisible by d .

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2.5: The Riemann Hypothesis

Definition 2.19

The Riemann zeta function $\zeta(z)$ is a complex function defined as follows on $\{z \in \mathbb{C} | \operatorname{Re} z > 1\}$

$$\zeta(z) = \sum_{n=1}^{\infty} n^{-z}$$

On other values of $z \in \mathbb{C}$ it is defined by the analytic continuation of this function (except at $z = 1$ where it has a simple pole).

Analytic continuation is akin to replacing e^x where x is real by e^z where z is complex. Another example is the series $\sum_{j=0}^{\infty} z^j$. This series diverges for $|z| > 1$. But as an analytic function, it can be replaced by $(1 - z)^{-1}$ on all of $\mathbb{C}$ except at the pole $z = 1$ where it diverges.

Recall that an analytic function is a function that is differentiable. Equivalently, it is a function that is locally given by a convergent power series. If f and g are two analytic continuations to a region U of a function h given on a region $V \subset U$, then the difference $f - g$ is zero on some U and therefore all its power expansions are zero and so it must be zero on the entire region. Hence, analytic conjugations are unique. That is the reason they are meaningful. For more details, see for example [4, 14].

It is customary to denote the argument of the zeta function by s . We will do so from here on out. Note that $|n - s| = n - \operatorname{Re} s$, and so for $\operatorname{Re} s > 1$ the series is absolutely convergent. At this point, the student should remember – or look up in [23] – the fact that absolutely convergent series can be rearranged arbitrarily without changing the sum. This leads to the following proposition.

Proposition 2.20

For $\operatorname{Re} s > 1$ we have

$$\sum_{n=1}^{\infty} n^{-s} = \prod_{p \text{ prime}} (1 - p^{-s})^{-1}$$

There are two common proofs of this formula. It is worth presenting both.

Proof

The first proof uses the Fundamental Theorem of Arithmetic. First, we recall that use geometric series

$$(1 - p^{-s})^{-1} = \sum_{k=0}^{\infty} p^{-ks}$$

to rewrite the right hand of the Euler product. This gives

$$\prod_{p \text{ prime}} (1 - p^{-s})^{-1} = \left(\sum_{k_1=0}^{\infty} p_1^{-k_1 s} \right) \left(\sum_{k_2=0}^{\infty} p_2^{-k_2 s} \right) \left(\sum_{k_3=0}^{\infty} p_3^{-k_3 s} \right)$$

Re-arranging terms yields

$$\dots = (p_1^{k_1} p_2^{k_2} p_3^{k_3} \dots)^{-s}$$

By the Fundamental Theorem of Arithmetic, the expression $(p_1^{k_1} p_2^{k_2} p_3^{k_3} \dots)$ runs through all positive integers exactly once. Thus upon re-arranging again we obtain the left hand of Euler's formula.

The second proof, the one that Euler used, employs a sieve method. This time, we start with the left hand of the Euler product. If we multiply ζ by 2^{-s} , we get back precisely the terms with n even. So

$$(1 - 2^{-s})\zeta(s) = 1 + 3^{-s} + 5^{-s} + \dots = \sum_{2 \nmid n} n^{-s}$$

Subsequently we multiply this expression by $(1 - 3^{-s})$. This has the effect of removing the terms that remain where n is a multiple of 3. It follows that eventually

$$(1 - p_1^{-s}) \dots (1 - p_l^{-s}) \zeta(s) = \sum_{p_1 \nmid n, \dots, p_l \nmid n} n^{-s}$$

The argument used in Eratosthenes sieve (Section 1.1) now serves to show that in the right hand side of the last equation all terms other than 1 disappear as l tends to infinity. Therefore, the left hand tends to 1, which implies the proposition.

The most important theorem concerning primes is probably the following (without proof).

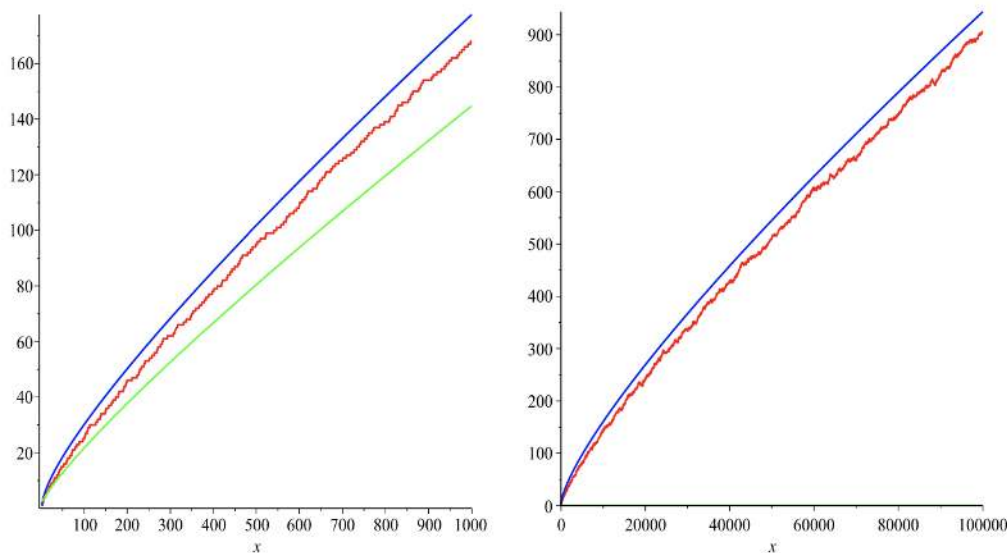


Figure 3. On the left, the function $\int_2^x \ln t dt$ in blue, $\pi(x)$ in red, and $x/\ln x$ in green. On the right, we have $\int_2^x \ln t dt - x/\ln x$ in blue, $\pi(x) - x/\ln x$ in red.

Theorem 2.21 (Prime Number Theorem)

Let $\pi(x)$ denote the prime counting function, that is: the number of primes less than or equal to $x > 2$.

Then

1. $\lim_{x \rightarrow \infty} \frac{\pi(x)}{(x/\ln x)} = 1$ and
2. $\lim_{x \rightarrow \infty} \frac{\pi(x)}{\int_2^x \ln t dt} = 1$

where $\ln$ is the natural logarithm.

The first estimate is the easier one to prove, the second is the more accurate one. In Figure 3 on the left, we plotted, for $x \in [2, 1000]$ from top to bottom the functions $\int_2^x \ln t dt$ in blue, $\pi(x)$ in red, and $x/\ln x$. In the right hand figure, we augment the domain to $x \in [2, 100000]$ and plot the difference of these functions with $x/\ln x$. It now becomes clear that $\int_2^x \ln t dt$ is indeed a much better approximation of $\pi(x)$. From this figure one may be tempted to conclude that $\int_2^x \ln t dt - \pi(x)$ is always greater than or equal to zero. This, however, is false. It is known that there are infinitely many n for which $\int_2^x \ln t dt - \pi(x) < 0$. The first such n is called the Skewes number. Not much is known about this number, except that it is less than 10317.

Perhaps the most important open problem in all of mathematics is the following. It concerns the analytic continuation of $\zeta(s)$ given above.

Conjecture 2.22 (Riemann Hypothesis)

All non-real zeros of $\zeta(s)$ lie on the line $\text{Re } s = \frac{1}{2}$

In his only paper on number theory [20], Riemann realized that the hypothesis enabled him to describe detailed properties of the distribution of primes in terms of the location of the non-real zero of $\zeta(s)$. This completely unexpected connection

between so disparate fields – analytic functions and primes in $\mathbb{N}$ —spoke to the imagination and led to an enormous interest in the subject. In further research, it has been shown that the hypothesis is also related to other areas of mathematics, such as, for example, the spacings between eigenvalues of random Hermitian matrices [2], and even physics [5, 6].

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2.6: Exercises

Exercise 2.6.1

Apply the division algorithm to the following number pairs. (Hint: replace negative numbers by positive ones.)

1. 110, 7.
2. 51, -30.
3. -138, 24
4. 272, 119
5. 2378, 1769
6. 270, 175560

Exercise 2.6.2

In this exercise we will exhibit the division algorithm applied to polynomials $x + 1$ and $3x^3 + 2x + 1$ with coefficients in $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$.

1. Apply long division to divide 3021 by 11. (Hint: $3021 = 11 \cdot 275 - 4$.)
2. Apply the exact same algorithm to divide $3x^3 + 2x + 1$ by $x + 1$. In this algorithm, x^k behaves as 10^k in (a). (Hint: every step, cancel the highest power of x .)
3. Verify that you obtain $3x^3 + 2x + 1 = (x + 1)(3x^2 - 3x + 5) - 4$.
4. Show that in general, if p_1 and p_2 are polynomials such that the degree of p_1 is greater or equal to the degree of p_2 , then $p_1 = q_2 p_2 + p_3$, where the degree of p_3 is less than the degree of p_2 . (Hint: perform long division as in (b). Stop when the degree of the remainder is less than that of p_2 .)
5. Why does this division not work for polynomials with coefficients in $\mathbb{Z}$? (Hint: replace $x + 1$ by $2x + 1$.)

Exercise 2.6.3

1. Compute by long division that $3021 = 11 \cdot 274 + 7$.
2. Conclude from exercise 2.2 that $3021 = 11(300 - 30 + 5) - 4$. (Hint: let $x = 10$.)
3. Conclude from exercise 2.2 that $3 \cdot 163 + 2 \cdot 16 + 1 = 17(3 \cdot 162 - 3 \cdot 16 + 5) - 4$. (Hint: let $x = 16$.)

Exercise 2.6.4

1. Use unique factorization to show that any composite number n must have a prime factor less than or equal to $\sqrt{n}$.
2. Use that fact to prove: If we apply Eratosthenes' sieve to $\{2, 3, \dots, n\}$, it is sufficient to sieve out numbers less than or equal to $\sqrt{n}$.

Exercise 2.6.5

We give an elementary proof of Corollary 2.15.

1. Show that $a \cdot \frac{b}{\gcd(a,b)}$ is a multiple of a .
2. Show that $\frac{b}{\gcd(a,b)} \cdot b$ is a multiple of b .
3. Conclude that $\frac{ab}{\gcd(a,b)}$ is a multiple of both a and b and thus greater than or equal to $\text{lcm}(a, b)$.
4. Show that $a / (\frac{ab}{\text{lcm}(a,b)}) = \frac{\text{lcm}(a,b)}{b}$ is an integer. Thus $\frac{ab}{\text{lcm}(a,b)}$ is a divisor of a .
5. Similarly, show that $\frac{ab}{\text{lcm}(a,b)}$ is a divisor of b .
6. Conclude that $\frac{ab}{\text{lcm}(a,b)} \leq \gcd(a, b)$.
7. Finish the proof.

Exercise 2.6.6

It is possible to extend the definition of gcd and lcm to more than two integers (not all of which are zero). For example $\gcd(24, 27, 54) = 3$

1. Compute $\gcd(6, 10, 15)$ and $\text{lcm}(6, 10, 15)$
2. Give an example of a triple whose gcd is one, but every pair of which has a gcd greater than one.
3. Show that there is no triple $\{a, b, c\}$ whose lcm equals abc , but every pair of which has lcm less than the product of that pair. (Hint: consider $\text{lcm}(a \cdot b) \cdot c$.)

Exercise 2.6.7

1. Give the prime factorization of the following numbers: 12, 392, 1043, 31, 128, 2160, 487
2. Give the prime factorization of the following numbers: $12 \cdot 392$, $1043 \cdot 31$, $128 \cdot 2160$
3. Give the prime factorization of: 1, 250000, 63^3 , 720 and the product of the last three numbers.

Exercise 2.6.8

Use the Fundamental Theorem of Arithmetic to prove:

1. Bézout's Lemma.
2. Euclid's Lemma.

Exercise 2.6.9

For positive integers m and n , suppose that $m^\alpha = n$. Show that $\alpha = \frac{p}{q}$ with $\gcd(p, q) = 1$ if and only if

$$m = \prod_{i=1}^s p_i^{k_i} \quad \text{and} \quad n = \prod_{i=1}^s p_i^{l_i} \quad \text{with} \quad \forall i, p k_i = q l_i$$

Exercise 2.6.10

We develop the proof of Theorem 2.16 as it was given by Euler. We start by assuming that there is a finite list L of k primes. We will show in the following steps how that assumption leads to a contradiction. We order the list according to ascending order of magnitude of the primes. So $L = \{p_1, p_2, \dots, p_k\}$ where $p_1 = 2, p_2 = 3, p_3 = 5$, and so forth, upto the last prime p_k .

1. Show that $\prod_{i=1}^k \frac{p_i}{p_i - 1}$ is finite, say M .
2. Show that for $r > 0$,

$$\prod_{i=1}^k \frac{p_i}{p_i - 1} = \prod_{i=1}^k \frac{1}{1 - p_i^{-1}} > \prod_{i=1}^k \frac{1 - p_i^{-r-1}}{1 - p_i^{-1}} = \prod_{i=1}^k \left(\sum_{j=0}^r p_i^{-j} \right)$$

3. Use the fundamental theorem of arithmetic to show that there is an $\alpha(r) > 0$ such that

$$\prod_{i=1}^k \left(\sum_{j=0}^r p_i^{-j} \right) = \sum_{l=1}^{\alpha(r)} \frac{1}{l} + R$$

, where R is a non-negative remainder.

4. Show that for all K there is an r such that $\alpha(r) > K$.
5. Thus for any K , there is an r such that

$$\prod_{i=1}^k \left(\sum_{j=0}^r p_i^{-j} \right) \geq \sum_{l=1}^K \frac{1}{l}$$

6. Conclude with a contradiction between a) and e). (Hint: the harmonic series $\sum \frac{1}{l}$ diverges or see exercise 2.11 c).)

Exercise 2.6.11

In this exercise we consider the Riemann zeta function for real values of s greater than 1.

1. Show that for all $x > -1$, we have $\ln(1+x) \leq x$.
2. Use Proposition 2.20 and a) to show that

$$\ln \zeta(s) \leq \sum_{p \text{ prime}} \frac{p^{-s}}{1-p^{-s}} \leq \sum_{p \text{ prime}} \frac{p^{-s}}{1-2^{-s}}$$

3. Use the following argument to show that $\lim_{s \rightarrow 1} \zeta(s) = \infty$.

$$\sum_{n=1}^{\infty} n^{-1} \geq \sum_{n=1}^{\infty} n^{-s} \geq \int_1^{\infty} x^{-s} dx$$

(For the last inequality, see Figure 4.)

4. Show that b) and c) imply that $\sum_{p \text{ prime}} p^{-s}$ diverges as $s \rightarrow 1$.
5. Show that – in some sense – primes are more frequent than squares in the natural numbers. (Hint: $\sum_{n=1}^{\infty} n^{-2}$ converges.)

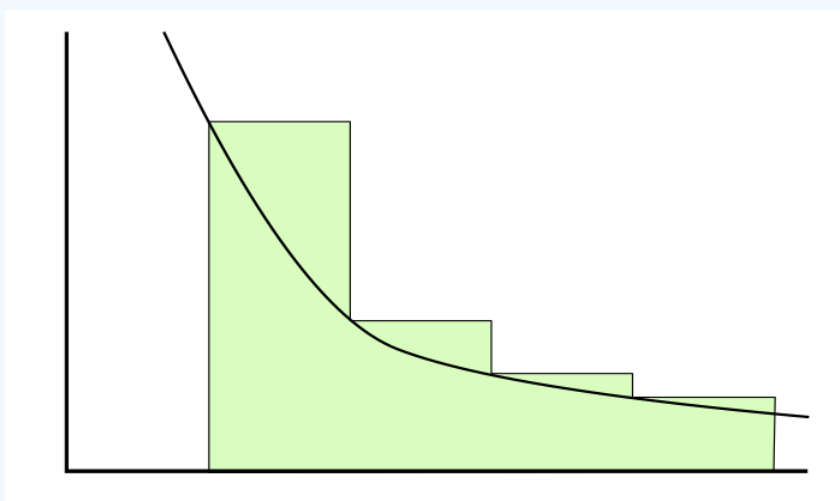


Figure 4. Proof that $\sum_{n=1}^{\infty} f(n)$ is greater than $\int_1^{\infty} x^{-s} dx$ if f is positive and decreasing.

Exercise 2.6.12

Let E be the set of even numbers. Let a, c in E , then c is divisible by a if there is $ab \in E$ so that $ab = c$. Define a prime p in E as a number in E such that there are no a and b in E with $ab = p$.

1. List the first 30 primes in E .
2. Does Euclid's lemma hold in E ? Explain.
3. Factor 60 into primes (in E) in two different ways.

Exercise 2.6.13

See exercise 2.12. Show that any number in E is a product of primes in E . (Hint: follow the proof of Theorem 2.14, part (1).)

Exercise 2.6.14

See exercise 2.12 which shows that unique factorization does not hold in $E = \{2, 4, 6, \dots\}$. The proof of unique factorization uses Euclid's lemma. In turn, Euclid's lemma was a corollary of Bézout's lemma, which depends on the division algorithm. Where exactly does the chain break down in this case?

Exercise 2.6.15

Let $L = \{p_1, p_2, \dots\}$ be the list of all primes, ordered according ascending magnitude. Show that $p_{n+1} \leq \prod_{i=1}^n p_i$. (Hint: consider d as in the proof of Theorem 2.16, let p_{n+1} be the smallest prime divisor of $d - 1$.)

A much stronger version of exercise 2.15 is the so-called Bertrand's Postulate. That theorem says that for every $n \geq 1$, there is a prime in $\{n + 1, \dots, 2n\}$. It was proved by Chebyshev. Subsequently the proof was simplified by Ramanujan and Erdős [18].

Exercise 2.6.16

Let p and q primes greater than or equal to 5.

1. Show that $p = 12q + r$ where $r \in \{1, 5, 7, 11\}$. (The same holds for q .)
2. Show that $24 \mid p^2 - q^2$. (Hint: use (a) to reduce this to $\binom{4}{2} = 6$ cases.)

Exercise 2.6.17

A square full number is a number $n > 1$ such that each prime factor occurs with a power at least 2. A square free number is a number $n > 1$ such that each prime factor occurs with a power at most 1.

1. If n is square full, show that there are positive integers a and b such that $n = a^2 b^3$.
2. Show that every integer greater than one is the product of a square free number and a square number.

Exercise 2.6.18

Let $L = \{p_1, p_2, \dots\}$ be the list of all primes, ordered according ascending magnitude. The numbers $E_n = 1 + \prod_{i=1}^n p_i$ are called Euclid numbers.

1. Check the primality of E_1 through E_6 .
2. Show that $E_n \equiv 3 \pmod{4}$. (Hint: $E_n - 1$ is twice an odd number.)
3. Show that for $n \leq 3$ the decimal representation of E_n ends in a 1. (Hint: look at the factors of E_n .)

Exercise 2.6.19

Twin primes are a pair of primes of the form p and $p + 2$.

1. Show that the product of two twin primes plus one is a square.
2. Show that $p > 3$, the sum of twin primes is divisible by 12. (Hint: see exercise 2.16)

Exercise 2.6.20

Show that there are arbitrarily large gaps between successive primes. More precisely, show that every integer in $\{n! + 2, n! + 3, \dots, n! + n\}$ is composite.

The usual statement for the fundamental theorem of arithmetic includes only natural numbers $n \in \mathbb{N}$ (i.e. not $\mathbb{Z}$) and the common proof uses induction on n . We review that proof in the next two problems.

Exercise 2.6.21

1. Prove that 2 can be written as a product of primes.
2. Let $k > 2$. Suppose all numbers in $\{1, 2, \dots, k\}$ can be written as a product of primes (or 1). Show that $k + 1$ is either prime or composite.
3. If in (b), $k + 1$ is prime, then all numbers in $\{1, 2, \dots, k + 1\}$ can be written as a product of primes (or 1).
4. If in (b), $k + 1$ is composite, then there is a divisor $d \in \{2, \dots, k\}$ such that $k + 1 = dd'$.
5. Show that the hypothesis in (b) implies also in this case, all numbers in $\{1, 2, \dots, k + 1\}$ can be written as a product of primes (or 1).
6. Use the above to formulate the inductive proof that all elements of $\mathbb{N}$ can be written as a product of primes.

Exercise 2.6.22

The set-up of the proof is the same as in exercise 2.21. Use induction on n . We assume the result of that exercise.

1. Show that $n = 2$ has a unique factorization.
2. Suppose that if for $k > 2$, $\{2, \dots, k\}$ can be uniquely factored. Then there are primes p_i and q_i , not necessarily distinct, such that

$$k + 1 = \prod_{i=1}^s p_i = \prod_{i=1}^r q_i$$

3. Show that then p_1 divides $\prod_{i=1}^r q_i$ and so, Corollary 2.12 implies that there is a $j \leq r$ such that $p_1 = q_j$.
4. Relabel the q_i 's, so that $p_1 = q_1$ and divide n by $p_1 = q_1$. Show that

$$\frac{k+1}{q_1} = \prod_{i=2}^s p_i = \prod_{i=2}^r q_i$$

5. Show that the hypothesis in (b) implies that the remaining p_i equal the remaining q_i . (Hint: $\frac{k}{q_1} \leq k$.)
6. Use the above to formulate the inductive proof that all elements of $\mathbb{N}$ can be uniquely factored as a product of primes.

Here is a different characterization of gcd and lcm. We prove it as a corollary of the prime factorization theorem.

Corollary 2.23

1. A common divisor $d > 0$ of a and b equals $\gcd(a, b)$ if and only if every common divisor of a and b is a divisor of d .
2. Also, a common multiple $d > 0$ of a and b equals $\text{lcm}(a, b)$ if and only if every common multiple of a and b is a multiple of d .

Exercise 2.6.23

Use the characterization of $\gcd(a, b)$ and $\text{lcm}(a, b)$ given in the proof of Corollary 2.15 to prove Corollary 2.23.

Exercise 2.6.24

1. Let p be a fixed prime. Show that the probability that two independently chosen integers in $\{1, 2, \dots, n\}$ are divisible by p tends to $\frac{1}{p^2}$ as $n \rightarrow \infty$. Equivalently, the probability that they are not divisible by p tends to $1 - \frac{1}{p^2}$.
2. Make the necessary assumptions, and show that the probability that two independently chosen integers in $\{1, 2, \dots, n\}$ are not divisible by any prime tends to $\prod_{p \text{ prime}} (1 - p^{-2})$. (Hint: you need to assume that the probabilities in 1. are independent and so they can be multiplied.)
3. Show that from 2. and Euler's product formula, it follows that for 2 random (positive) integers a and b to have $\gcd(a, b) = 1$ has probability $1/\zeta(2) \approx 0.61$
4. Show that $d > 1$ integers $\{a_1, a_2, \dots, a_d\}$ that probability equals $1/\zeta(d)$. (Hint: the reasoning is the same as in 1., 2., 3.)
5. Show that for $d \geq 2$:

$$1 < \zeta(d) < 1 + \int_1^\infty x^{-d} dx$$

For the last inequality, see Figure 5.

6. Show that for large d , the probability that $\gcd(a_1, a_2, \dots, a_d) = 1$ tends to 1.

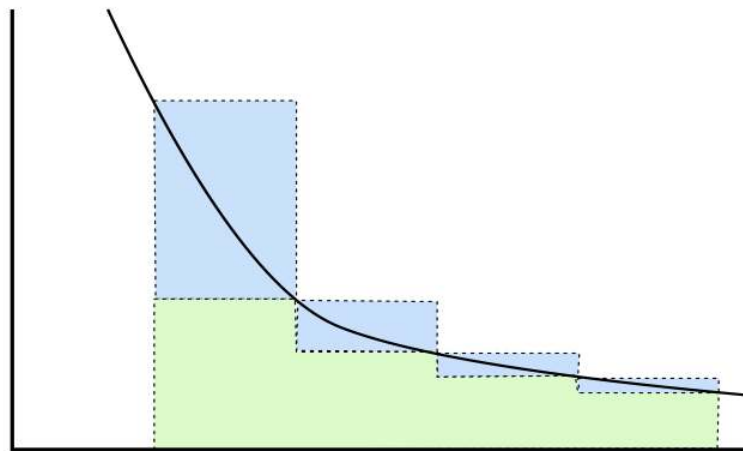


Figure 5. Proof that $\sum_{n=1}^{\infty} f(n)$ (shaded) minus $f(1)$ (shaded in blue) is less than $\int_1^{\infty} x^{-s} dx$ is positive and decreasing.

Exercise 2.6.25

This exercise is based on exercise 2.24.

1. In the $\{-4, \dots, 4\}^2 \setminus (0, 0)$ grid in $\mathbb{Z}^2$, find out which proportion of the lattice points is visible from the origin, see Figure 6.
2. Show that in a large grid, this proportion tends to $1/\zeta(2)$.
3. Show that as the dimension increases to infinity, the proportion of the lattice points $\mathbb{Z}^d$ that are visible from the origin, increases to 1.

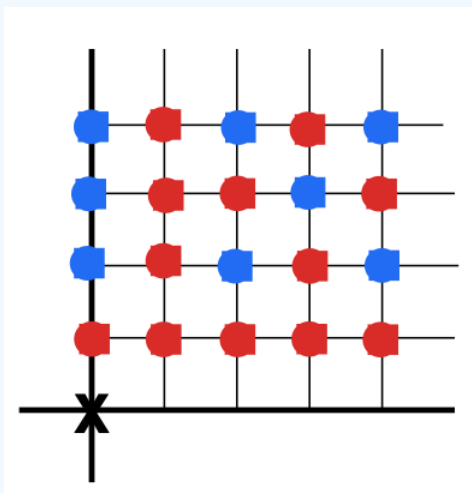


Figure 6. The origin is marked by “ $\times$ ”. The red dots are visible from $\times$; between any blue dot and $\times$ there is a red dot. The picture shows exactly one quarter of $\{-4, \dots, 4\}^2 \setminus (0, 0) \subset \mathbb{Z}^2$.

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CHAPTER OVERVIEW

3: Linear Diophantine Equations

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3.1: The Euclidean Algorithm

Lemma 3.1

In the division algorithm of Definition 2.4, we have $\gcd(r_1, r_2) = \gcd(r_2, r_3)$.

Proof

On the one hand, we have $r_1 = r_2 q_2 + r_3$, and so any common divisor of r_2 and r_3 must also be a divisor of r_1 (and of r_2). Vice versa, since $r_1 - r_2 q_2 = r_3$, we have that any common divisor of r_1 and r_2 must also be a divisor of r_3 (and of r_2).

Thus by calculating r_3 , the residue of r_1 modulo r_2 , we have simplified the computation of $\gcd(r_1, r_2)$. This is because r_3 is strictly smaller (in absolute value) than both r_1 and r_2 . In turn, the computation of $\gcd(r_2, r_3)$ can be simplified similarly, and so the process can be repeated. Since the r_i form a monotone decreasing sequence in $\mathbb{N}$, this process must end when $r_{n+1} = 0$ after a finite number of steps. We then have $\gcd(r_1, r_2) = \gcd(r_n, 0) = r_n$.

Corollary 3.2

Given $r_1 > r_2 > 0$, apply the division algorithm until $r_n > r_{n+1} = 0$. Then $\gcd(r_1, r_2) = \gcd(r_n, 0) = r_n$. Since r_i is decreasing, the algorithm always ends.

Definition 3.3

The repeated application of the division algorithm to compute $\gcd(r_1, r_2)$ is called the Euclidean algorithm.

We now give a framework to reduce the messiness of these repeated computations. Suppose we want to compute $\gcd(188, 158)$. We do the following computations:

$$188 = 158 \cdot 1 + 30$$

$$158 = 30 \cdot 5 + 8$$

$$30 = 8 \cdot 3 + 6$$

$$8 = 6 \cdot 1 + 2$$

$$6 = 2 \cdot 3 + 0$$

We see that $\gcd(188, 158) = 2$. The numbers that multiply the r_i are the quotients of the division algorithm (see the proof of Lemma 2.3). If we call them q_{i-1} , the computation looks as follows:

$$\begin{aligned} r_1 &= r_2 q_2 + r_3 \\ r_2 &= r_3 q_3 + r_4 \\ &\vdots \\ r_{n-3} &= r_{n-2} q_{n-2} + r_{n-3} \\ r_{n-2} &= r_{n-1} q_{n-1} + r_{n-2} \\ r_{n-1} &= r_n q_n + 0 \end{aligned} \tag{3.1.1}$$

where we use the convention that $r_{n+1} = 0$ while $r_n \neq 0$. Observe that with that convention, (3.1) consists of $n - 1$ steps. A much more concise form (in part based on a suggestion of Katahdin [16]) to render this computation is as follows.

	q_n	q_{n-1}	$\cdots$	q_3	q_2	
0	r_n	r_{n-1}	$\cdots$	r_3	r_2	r_1

Thus, each step $r_{i+1} | r_i$ is similar to the usual long division, except that its quotient q_{i+1} is placed above r_{i+1} (and not above r_i), while its remainder r_{i+2} is placed all the way to the left of r_{i+1} . The example we worked out before now it looks like:

$$\begin{array}{r|cccccc}
 & 3 & 1 & 3 & 5 & 1 & \\
 \hline
 0 & 2 & 6 & 8 & 30 & 158 & 188
 \end{array}$$

There is a beautiful visualization of this process outlined in exercise 3.4.

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3.2: A Particular Solution of $ax+by = c$

Another interesting way to encode the computations done in equations 3.1 and 3.2 is via matrices.

$$\begin{pmatrix} r_{i-1} \\ r_i \end{pmatrix} = \begin{pmatrix} q_i & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} r_i \\ r_{i+1} \end{pmatrix} \quad (3.2.1)$$

Denote the matrix in this equation by Q_i . Its determinant equals -1 , and so it is invertible. In fact,

$$Q_i = \begin{pmatrix} q_i & 1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad Q_i^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & -q_i \end{pmatrix} \quad (3.2.2)$$

These matrices Q_i are very interesting. We will use them again to study the theory of continued fractions in Chapter 6. For now, as we will see in Theorem 3.4, they give us an explicit algorithm to find a solution to the equation $r_1x + r_2y = r \gcd(r_1, r_2)$. Note that from Bezout's Lemma (Lemma 2.5), we already know this has a solution. But the next result gives us a simple way to actually calculate a solution. In what follows X_{ij} means the (i, j) entry of the matrix X .

Theorem 3.4

Give r_1 and r_2 , a solution for x and y of $r_1x + r_2y = r \gcd(r_1, r_2)$ is given by

$$x = r(Q_{n-1}^{-1} \cdots Q_2^{-1})_{2,1} \quad \text{and} \quad y = r(Q_{n-1}^{-1} \cdots Q_2^{-1})_{2,2}$$

Proof

Let r_i, q_i , and Q_i be defined as above, and set $r_{n+1} = 0$. From equation 3.4, we have

$$\begin{pmatrix} r_i \\ r_{i+1} \end{pmatrix} = Q_i^{-1} \begin{pmatrix} r_{i-1} \\ r_i \end{pmatrix} \Rightarrow r \begin{pmatrix} r_{n-1} \\ r_n \end{pmatrix} = r Q_{n-1}^{-1} \cdots Q_2^{-1} \begin{pmatrix} r_1 \\ r_2 \end{pmatrix}$$

Observe that $r_{n+1} = 0$ and so $\gcd(r_1, r_2) = r_n$ and

$$\begin{pmatrix} r_{n-1} \\ r_n \end{pmatrix} = \begin{pmatrix} x_{n-1} & y_{n-1} \\ x_n & y_n \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \end{pmatrix} \quad (3.2.3)$$

The equalities follow immediately.

In practice, rather than multiplying all these matrices, it may be more convenient to solve equation 3.1 or 3.2 “backward”, as the expression goes. This can be done as follows. Start with

$$\gcd(r_1, r_2) = r_n = r_{n-2} - r_{n-1}q_{n-1}$$

which follows from equation 3.1. The line above it in that same equation gives $r_{n-1} = r_{n-3} - r_{n-2}q_{n-2}$. Use this to eliminate r_{n-1} in favor of r_{n-2} and r_{n-3} . So,

$$\begin{aligned} \gcd(r_1, r_2) &= r_n = r_{n-2} - (r_{n-3} - r_{n-2}q_{n-2})q_{n-1} \\ &= r_{n-2}(1 + q_{n-1}q_{n-2}) + r_{n-3}(-q_{n-1}) \end{aligned}$$

This computation can be done still more efficiently by employing the notation of equation 3.2 again.

	+	−	+	−	+	
	q_n	q_{n-1}	q_{n-2}	q_{n-3}	q_{n-4}	
0	r_n	r_{n-1}	r_{n-2}	r_{n-3}	r_{n-4}	...
	1					
	0	$-q_{n-1}$	1			
			$q_{n-1}q_{n-2}$	$-q_{n-1}$		
				$-q_{n-3}(1 + q_{n-1}q_{n-2})$	$1 + q_{n-1}q_{n-2}$	

(The signs added in the first line in this scheme serve only to keep track of the signs of the coefficients in lines three and below.)
Applying this to the example gives

	+	−	+	−	+	−
	3	1	3	5	1	
0	2	6	8	30	158	188
	1					
		−1				
			$\frac{1}{3}$			
				$-\frac{1}{20}$		
					$\frac{4}{21}$	
						−21

Adding the last two lines gives that $2 = 158(25) + 188(-21)$.

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3.3: Solution of the Homogeneous equation $ax+by = 0$

Proposition 3.5

The general solution of the homogeneous equation $r_1x + r_2y = 0$ is given by

$$x = k \frac{r_2}{\gcd(r_1, r_2)} \quad \text{and} \quad y = -k \frac{r_1}{\gcd(r_1, r_2)}$$

where $k \in \mathbb{Z}$.

Proof

On the one hand, by substitution the expressions for x and y into the homogeneous equation, one checks they are indeed solutions. On the other hand, x and y must satisfy

$$\frac{r_1}{\gcd(r_1, r_2)}x = -\frac{r_2}{\gcd(r_1, r_2)}y$$

The integers r_i (for i in $\{1, 2\}$) have greatest common divisor equal $\gcd(r_1, r_2)$ to 1. Thus Euclid's lemma applies and therefore $\frac{r_1}{\gcd(r_1, r_2)}$ is a divisor of y , while $\frac{r_2}{\gcd(r_1, r_2)}$ is a divisor of x .

A different proof of this lemma goes as follows. The set of all solution in $\mathbb{R}^2$ of $r_1x + r_2y = 0$ is given by the line $l(\xi) = \begin{pmatrix} r_2 \\ -r_1 \end{pmatrix} \xi$. To obtain all its lattice points (i.e., points that are also in $\mathbb{Z}^2$), both $r_2\xi$ and $-r_1\xi$ must be integers. The smallest positive number ξ for which this is possible, is

$$\xi = \frac{1}{\gcd(r_1, r_2)}$$

Here is another homogeneous problem that we will run into. First we need a small update of Definition 1.2.

Definition 3.6

Let $\{b_i\}_{i=1}^n$ be non-zero integers. Their greatest common divisor, $\gcd(b_1, \dots, b_n)$, is the maximum of the numbers that are divisors of every b_i ; their least common multiple, $\text{lcm}(b_1, \dots, b_n)$, is the least of the positive numbers that are multiples of every b_i .

Surprisingly, for this more general definition, the generalization of Corollary 2.15 is false. For an example, see exercise 2.6.

Corollary 3.7

Let $\{b_i\}_{i=1}^n$ be non-zero integers and denote $B = \text{lcm}(b_1, \dots, b_n)$. The general solution of the homogeneous system of equations $x =_{b_i} 0$ is given by

$$x =_B 0$$

Proof

From the definition of $\text{lcm}(b_1, \dots, b_n)$, every such x is a solution. On the other hand, if $x \neq_B 0$, then there is an i such that x is not a multiple of b_i , and therefore such an x is not a solution.

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3.4: The General Solution of $ax+by = c$

Definition 3.8

Let r_1 and r_2 be given. The equation $r_1x + r_2y = 0$ is called homogeneous. The equation $r_1x + r_2y = c$ when $c \neq 0$ is called inhomogeneous. An arbitrary solution of the inhomogeneous equation is called a particular solution. By general solution, we mean the set of all possible solutions of the full (inhomogeneous) equation.

It is useful to have some geometric intuition relevant to the equation $r_1x + r_2y = c$. In $\mathbb{R}^2$, we set $\vec{r} = (r_1, r_2)$, $\vec{x} = (x, y)$, etcetera. The standard inner product is written as $(\cdot, \cdot)$. The set of points in $\mathbb{R}^2$ satisfying the above inhomogeneous equation thus lie on the line $m \subset \mathbb{R}^2$ given by $(\vec{r}, \vec{x}) = c$.

This line is orthogonal to the vector $\vec{r}$ and its distance to the origin (measured along the vector $\vec{r}$) equals $\frac{|c|}{\sqrt{(\vec{r}, \vec{r})}}$. The situation is illustrated in Figure 7.

It is a standard result from linear algebra that the problem of finding all solutions of a inhomogeneous equation comes down to finding one solution of the inhomogeneous equation, and finding the general solution of the homogeneous equation.

Lemma 3.9

Let $(x^{(0)}, y^{(0)})$ be a particular solution of $r_1x + r_2y = c$. The general solution of the inhomogeneous equation is given by $(x^{(0)} + z_1, y^{(0)} + z_2)$ where (z_1, z_2) is the general solution of the homogeneous equation $r_1x + r_2y = 0$.

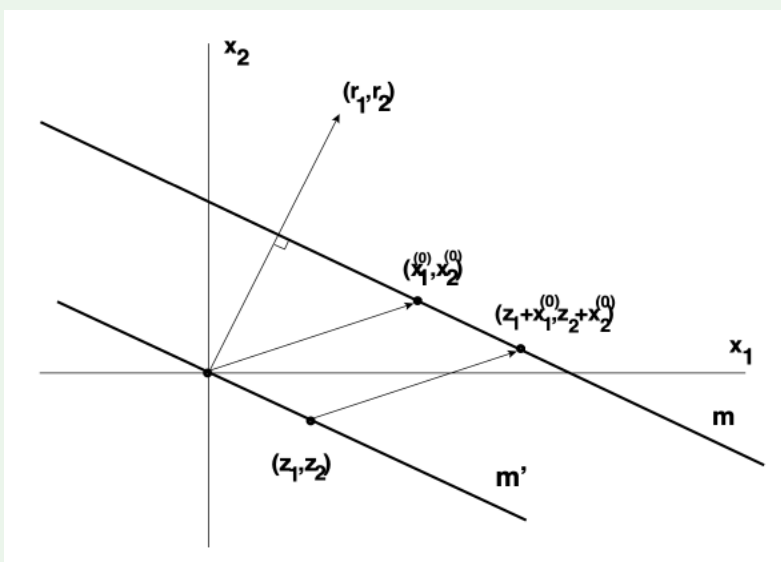


Figure 7. The general solution of the inhomogeneous equation $(\vec{r}, \vec{x}) = c$ in $\mathbb{R}^2$.

Proof

Let $(x^{(0)}, y^{(0)})$ be that particular solution. Let m be the line given by $(\vec{r}, \vec{x}) = c$. Translate m over the vector $(-x^{(0)}, -y^{(0)})$ to get the line m' . Then an integer point on the line m' is a solution (z_1, z_2) of the homogeneous equation if and only if $(x^{(0)} + z_1, y^{(0)} + z_2)$ on m is also an integer point (see Figure 7).

Be'zout's Lemma says that $r_1x + r_2y = c$ has a solution if and only if $\gcd(r_1, r_2) | c$. Theorem 3.4 gives a particular solution of that equation (via the Euclidean algorithm). Putting those results and Proposition 3.5 together, gives our final result.

Corollary 3.10

Given r_1, r_2 , and c , the general solution of the equation $r_1x + r_2y = c$, where $\gcd(r_1, r_2) | c$, is the sum of the particular solution of Theorem 3.4 and the general solution of $r_1x + r_2y = 0$ of Proposition 3.5.

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3.5: Recursive Solution of x and y in the Diophantine Equation

Theorem 3.4 has two interesting corollaries. The first is in fact stated in the proof of that theorem, and the second requires a very short proof. We will make extensive use of these two results in Chapter 6 when we discuss continued fractions.

Corollary 3.11

Given r_1, r_2 , and their successive remainders $r_3, \dots, r_n \neq 0$, and $r_{n+1} = 0$ in the Euclidean algorithm. Then for $i \in \{3, \dots, n\}$, the solution for (x_i, y_i) in $r_i = r_1 x_i + r_2 y_i$ is given by:

$$\begin{pmatrix} r_i \\ r_{i+1} \end{pmatrix} = Q_i^{-1} \dots Q_2^{-1} \begin{pmatrix} r_1 \\ r_2 \end{pmatrix}$$

Corollary 3.12

Given r_1, r_2 , and their successive quotients q_2 through q_n as in equation 3.1, then x_i and y_i of Corollary 3.11 can be solved as follows:

$$\begin{pmatrix} x_i & y_i \\ x_{i+1} & y_{i+1} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & -q_i \end{pmatrix} \begin{pmatrix} x_{i-1} & y_{i-1} \\ x_i & y_i \end{pmatrix} \quad \text{with} \quad \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Proof

The initial follows, because

$$r_1 = r_1 \cdot 1 + r_2 \cdot 0$$

$$r_2 = r_1 \cdot 0 + r_2 \cdot 1$$

Notice that, by definition,

$$\begin{pmatrix} r_i \\ r_{i+1} \end{pmatrix} = \begin{pmatrix} r_1 x_i + r_2 y_i \\ r_1 x_{i+1} + r_2 y_{i+1} \end{pmatrix} = \begin{pmatrix} x_i & y_i \\ x_{i+1} & y_{i+1} \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \end{pmatrix}$$

From Corollary 3.11, we now have that

$$\begin{pmatrix} r_i \\ r_{i+1} \end{pmatrix} = Q_i^{-1} \begin{pmatrix} r_{i-1} \\ r_i \end{pmatrix} \Rightarrow \begin{pmatrix} x_i & y_i \\ x_{i+1} & y_{i+1} \end{pmatrix} = Q_i^{-1} \begin{pmatrix} x_{i-1} & y_{i-1} \\ x_i & y_i \end{pmatrix}$$

From this, one deduces the equations for x_{i+1} and y_{i+1} .

We remark that the recursion in Corollary 3.12 can also be expressed as

$$x_{i+1} = -q_i x_i + x_{i-1}$$

$$y_{i+1} = -q_i y_i + y_{i-1}$$

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3.6 Exercises

Exercise 1

Which of the following equations have integer solutions?

1. $1137 = 69x + 39y$.
2. $1138 = 69x + 39y$.
3. $-64 = 147x + 84y$.
4. $-63 = 147x + 84y$.

Exercise 2

Let l be the line in $\mathbb{R}^2$ given by $y = \rho x$, where $\rho \in \mathbb{R}$.

1. Show that l intersects $\mathbb{Z}^2$ if and only if ρ is rational.
2. Given a rational $\rho > 0$, find the intersection of l with $\mathbb{Z}^2$. (Hint: set $\rho = r_1$ and use Proposition 3.5.)

Exercise 3

Apply the Euclidean algorithm to find the greatest common divisor of the following number pairs. (Hint: replace negative numbers by positive ones. For the division algorithm applied to these pairs, see exercise 2.1)

1. 110, 7.
2. 51, -30.
3. -138, 24
4. 272, 119
5. 2378, 1769
6. 270, 175, 560

Exercise 4

This problem was taken (and reformulated) from [1].

1. Tile a 188 by 158 rectangle by squares using what is called a greedy algorithm. The first square is 158 by 158. The remaining rectangle is 158 by 30. Now the optimal choice is five 30 by 30 squares. What remains is an 30 by 8 rectangle, and so on. Explain how this is a visualization of equation (3.3).
2. Consider equation (3.1) or (3.2) and use a) to show that

$$r_1 r_2 = \sum_{i=2}^n q_i r_i^2$$

(Hint: assume that $r_1 > r_2 > 0$, $r_n \neq 0$, and $r_{n+1} = 0$.)

Exercise 5

Determine if the following Diophantine equations admit a solution for x and y . If yes, find a (particular) solution. (Hint: Use one of the algorithms in Section 3.2.)

1. $110x + 7y = 13$.
2. $110x + 7y = 5$.
3. $51x - 30y = 6$.
4. $51x - 30y = 7$.
5. $-138x + 24y = 7$.
6. $-138x + 24y = 6$.
7. $272x + 119y = 54$.
8. $272x + 119y = 17$.

9. $2378x + 1769y = 300$
10. $2378x + 1769y = 57$
11. $270x + 175,560y = 170$
12. $270x + 175,560y = 150$

Exercise 6

Find the solutionsn for x and y of the following (homogeneous) Diophantine equations. If yes, find a (particular) solution. (Hint: Use one of the algorithms in Section 3.2.)

1. $110x + 7y = 0$.
2. $51x - 30y = 0$.
3. $-138x + 24y = 0$.
4. $272x + 119y = 0$.
5. $2378x + 1769y = 0$.
6. $270x + 175,560y = 0$.

Exercise 7

Find all solutions for x and y in all problems of exercise 3.5 that admit a solution.

Exercise 8

Use Corollary 3.11 to express the successive remainders r_i in each of the items in exercise 3.3 as $r_1x_i + r_2y_i$.

Exercise 9

Use Corollary 3.12 to express x_i and y_i in the successive remainders r_i in each of the items in exercise 3.3.

Exercise 10

Consider the line l in $\mathbb{R}$ defined by $l(\xi) = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix} \xi$, where $\xi \in \mathbb{R}$ and the r_i are integers.

1. Show that $l(\xi) \in \mathbb{Z}^3 \setminus \{\vec{0}\}$ if and only if $\xi = \frac{t}{\gcd(r_1, r_2, r_3)}$ and $t \in \mathbb{Z}$.
2. Show that this implies that if any of the r_i is irrational, then l has no non-zero points in common with $\mathbb{Z}^3$.

Exercise 11

For this exercise, see exercises 2.2 and 2.3. Given three polynomials p_1 and p_2 with coefficients in $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$. Define $\gcd(p_1, p_2)$ as the highest degree polynomial that is a divisor of both p_1 and p_2 . (Remark: if you do this very carefully, you will realize that in this problem you really need to consider equivalence classes of polynomials, in the sense that two polynomials p and q are equivalent if there is a non-zero constant $c \in \mathbb{R}$ such that $p = cq$. While we are not pursuing the details here, it is good to keep this in the back of your mind.)

1. Describe an algorithm based on the division algorithm for polynomials to determine the greatest common divisor $d(x)$ of $p_1(x)$ and $p_2(x)$. (Hint: see the Euclidean algorithm of Section 3.1. To see how to apply the division algorithm to polynomials, see exercise 2.2.)
2. Give a necessary and sufficient criterion for the existence of two polynomials $g(x)$ and $h(x)$ satisfying

$$p_1(x)g(x) + p_2(x)h(x) = d(x)$$

(Hint: the same as Be'zout's lemma.)

3. Apply the trick of "backward" solving in Section 3.2 to determine an algorithm find polynomials $g(x)$ and $h(x)$ satisfying

$$p_1(x)g(x) + p_2(x)h(x) = d(x)$$

where now $d = \gcd(p_1, p_2)$. This is the particular solution.

4. Find the general solution of

$$p_1(x)g(x) + p_2(x)h(x) = 0$$

(Hint: see Proposition 3.5.)

5. Find the general solution of

$$p_1(x)g(x) + p_2(x)h(x) = e(x)$$

where now e is a polynomial times $\gcd(p_1, p_2)$. (Hint: see Corollary 3.10.)

6. Apply the above to $p_1(x) = x^7 - x^2 + 1$, $p_2(x) = x^3 + x^2$, and $e(x) = 2 - x$. (Hint: We list the steps of the Euclidean algorithm:

$$(x^7 - x^2 + 1) = (x^3 + x^2)(x^4 - x^3 + x^2 - x + 1) + (-2x^2 + 1)$$

$$(x^3 + x^2) = (-2x^2 + 1)\left(-\frac{1}{2}x - \frac{1}{2}\right) + \left(\frac{1}{2}x + \frac{1}{2}\right)$$

$$(-2x^2 + 1) = \left(\frac{1}{2}x + \frac{1}{2}\right)(-4x + 4) + (-1)$$

$$\left(\frac{1}{2}x + \frac{1}{2}\right) = (-1)\left(-\frac{1}{2}x - \frac{1}{2}\right)$$

Thus $\gcd(p_1, p_2) = 1$, or in other words, $p_1(x)$ and $p_2(x)$ are relatively prime.)

Exercise 12

Let $p(x)$ be a polynomial and $p'(x)$ its derivative.

1. Show that if $p(x)$ has a multiple root λ of order $k > 1$, then $p'(x)$ has that same root of order $k - 1$. (Hint: Differentiate $p(x) = h(x)(x - \lambda)^k$.)
2. Use exercise 3.11, to give an algorithm to find a polynomial $q(x)$ that has the same roots as $p(x)$, but all roots are simple (i.e. no multiple roots). (Hint: you need to divide p by $\gcd(p, p')$.)

Theorem 3.13 (Fundamental Theorem of Algebra)

A polynomial in $\mathbb{C}[x]$ (the set of polynomials with complex coefficients) of degree $d \geq 1$ has exactly d roots, counting multiplicity.

Exercise 13

Assume that every polynomial f of degree $d \geq 1$ has at least 1 root, prove the fundamental theorem of algebra. (Hint: let ρ be a root and use the division algorithm to write $f(x) = (x - \rho)q(x) + r$ where r has degree 0.)

Exercise 14

Let f and g be polynomials in $\mathbb{Q}[x]$ with root ρ and suppose that g is minimal (Definition 1.13). Show that $g|f$. (Hint: use the division algorithm of exercises 2.2 and 2.3 to write $f(x) = g(x)q(x) + r(x)$ where r has degree less than g .)

Definition 3.14

The sequence $\{F_i\}_{i=0}^{\infty}$ of Fibonacci numbers F_i is defined as follows

$$F_0 = 0, \quad F_1 = 1, \quad \forall i > 1 : F_{i+1} = F_i + F_{i-1}$$

Exercise 15

Denote the golden mean, or $\frac{1+\sqrt{5}}{2} \approx 1.618$, by g .

1. Show that $g^2 = g + 1$ and thus $g^{n+1} = g^n + g^{n-1}$.
2. Show that $F_3 \geq g^1$ and $F_2 \geq g^0$.
3. Use induction to show that $F_{n+2} \geq g^n$ for $n > 0$.
4. Use the fact that $5 \log_{10} \frac{1+\sqrt{5}}{2} \approx 1.045$, to show that $F_{5k+2} > 10^k$ for $k \geq 0$.

Exercise 16

Consider the equations in (3.1) and assume that $r_{n+2} = 0$ and $r_{n+1} > 0$.

1. Show that $r_{n+1} \geq F_2 = 1$ and $r_n \geq F_3 = 2$. (Hint: $r(i)$ is strictly increasing.)
2. Show that $r_1 \geq F_{n+2}$.
3. Suppose r_1 and r_2 in $\mathbb{N}$ and $\max\{r_1, r_2\} < F_{n+2}$. Show that the Euclidean Algorithm to calculate $\gcd(r_1, r_2)$ takes at most $n - 1$ iterates of the division algorithm.

Exercise 17

Use exercises 3.15 and 3.16 to show that the Euclidean Algorithm to calculate $\gcd(r_1, r_2)$ takes at most $5k - 1$ iterates where k is the number of decimal places of $\max\{r_1, r_2\}$. (This is known as Lamé's theorem.)

Exercise 18

1. Write the numbers 287, 513, and 999 in base 2, 3, and 7, using the division algorithm. Do not use a calculating device. (Hint: start with base 10. For example,

$$287 = 28 \cdot 10 + 7$$

$$28 = 2 \cdot 10 + 8$$

$$2 = 0 \cdot 10 + 2$$

Hence the number of in base 10 is $2 \cdot 10^2 + 8 \cdot 10^1 + 3 \cdot 10^0$.)

2. Show that to write n in base b takes about $\log_b n$ divisions.

Exercise 19

Suppose $\{b_i\}_{i=1}^n$ are positive integers such that $\gcd(b_i, b_j) = 1$ for $i \neq j$. We want to know all z that satisfy

$$z = b_i c_i \quad \text{for} \quad i \in \{1, \dots, m\}$$

1. Set $B = \prod_{i=1}^n b_i$ and show that the homogeneous problem is solved by

$$z =_B 0$$

(Hint: use corollary 3.10)

2. Check that

$$z = \sum_{i=1}^n \frac{B}{b_i} x_i c_i$$

is a particular solution.

3. Find the general solution

$$z = B \sum_{i=1}^n \frac{B}{b_i} x_i c_i$$

This is known as the Chinese Remainder Theorem.

Exercise 20

Use exercise 3.19 to solve

$$z \equiv_2 1$$

$$z \equiv_3 2$$

$$z \equiv_5 3$$

$$z \equiv_7 5$$

Exercise 21

Assume $\gcd(F_n, F_{n+1}) = 1$. Use Exercise 3.19 to solve:

$$z \equiv_{F_n} F_{n-1}$$

$$z \equiv_{F_{n+1}} F_n$$

where F_n are the Fibonacci numbers of Definition 3.14.

Exercise 22 (The Chinese remainder theorem generalized)

Suppose $\{b_i\}_{i=1}^n$ are positive integers. We want to know all z that satisfy

$$z \equiv b_i c_i \quad \text{for } i \in \{1, \dots, m\}$$

1. Set $B = \text{lcm}(b_1, b_2, \dots, b_n)$ and show that the homogeneous problem is solved by

$$z \equiv_B 0$$

2. Show that if there is a particular solution then

$$\forall i \neq j: c_i \equiv_{\gcd(b_i, b_j)} c_j$$

3. Formulate the general solution when the condition in b) holds.

Exercise 23

Assume $\gcd(F_n, F_{n+1}) = 1$. Use Exercise 3.21 to solve:

$$z \equiv_6 15$$

$$z \equiv_{10} 6$$

$$z \equiv_{15} 10$$

See also exercise 2.6.

CHAPTER OVERVIEW

4: Number Theoretic Functions

[4.1: Multiplicative Functions](#)

[4.2: Additive Functions](#)

[4.3: Mobius inversion](#)

[4.4: Euler's Phi or Totient Function](#)

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4.1: Multiplicative Functions

Definition 4.1

Number theoretic functions or arithmetic functions or sequences are functions defined on the positive integers (i.e. $\mathbb{N}$) with values in $\mathbb{C}$.

Note that in other areas of mathematics, the word sequence is the only term commonly used. We will use these terms interchangeably.

Definition 4.2

A multiplicative function is a sequence such that $\gcd(a, b) = 1$ implies $f(ab) = f(a)f(b)$. A completely multiplicative function is one where the condition that $\gcd(a, b) = 1$ is not needed.

Note that completely multiplicative implies multiplicative (but not vice versa). The reason this definition is interesting, is that it allows us to evaluate the value of a multiplicative function f on any integer as long as we can compute $f(p^k)$ for any prime p . Indeed, using the fundamental theorem of arithmetic,

$$\text{if } n = \prod_{i=1}^r p_i^{l_i} \quad \text{then} \quad f(n) = \prod_{i=1}^r f(p_i^{l_i})$$

as follows immediately from Definition 4.2.

Proposition 4.3

Let f be a multiplicative function on the integers. Then

$$F(n) = \sum_{d|n} f(d)$$

is also multiplicative.

Proof

Let $n = \prod_{i=1}^r p_i^{l_i}$. The summation $\sum_{d|n} f(d)$ can be written out using the previous lemma and the fact that f is multiplicative:

$$\begin{aligned} F(n) &= \sum_{a_1=0}^{l_1} \cdots \sum_{a_r=0}^{l_r} f(p_1^{a_1}) \cdots f(p_r^{a_r}) \\ &= \prod_{i=1}^r \left(\sum_{a_i=0}^{l_i} f(p_i^{a_i}) \right) \end{aligned}$$

Now let a and b two integers greater than 1 and such that $\gcd(a, b) = 1$ and $ab = n$. Then by the Unique Factorization Theorem a and b can be written as:

$$a = \prod_{i=1}^r p_i^{l_i} \quad \text{and} \quad b = \prod_{i=r+1}^s p_i^{l_i}$$

Applying the previous computation to a and b yields that $f(a)f(b) = f(n)$.

Perhaps the simplest multiplicative functions are the ones where $f(n) = n^k$ for some fixed k . Indeed, $f(n)f(m) = n^k m^k = f(nm)$. In fact, this is a completely multiplicative function.

Definition 4.4

Let $k \in \mathbb{R}$. The multiplicative function $\sigma_k : \mathbb{N} \rightarrow \mathbb{R}$ gives the sum of the k -th power of the positive divisors of n . Equivalently:

$$\sigma_k(n) = \sum_{d|n} d^k$$

Note that the multiplicativity of σ_k follows directly from Proposition 4.3. Special cases are when $k = 1$ and $k = 0$. In the first case, the function is simply the sum of the positive divisors and the subscript '1' is usually dropped. When $k = 0$, the function is usually called τ , and the function's value is the number of positive divisors of its argument.

Theorem 4.5

Let $n = \prod_{i=1}^r p_i^{l_i}$ where p_i are primes. Then for $k \neq 0$

$$\sigma_k(n) = \prod_{i=1}^r \left(\frac{p_i^{k(l_i+1)} - 1}{p_i^k - 1} \right)$$

while for $k = 0$

$$\sigma_0(n) = \tau(n) = \prod_{i=1}^r (l_i + 1)$$

Proof

By Proposition 4.3, $\sigma_k(n)$ is multiplicative, so it is sufficient to compute for some prime p

$$\sigma_k(p^l) = \sum_{i=0}^l p_i^{ik} = \frac{p^{k(l+1)} - 1}{p^k - 1}$$

Thus, $\sigma_k(n)$ is indeed a product of these terms.

However, there are other interesting multiplicative functions beside the powers of the divisors. The Mobius function defined below is one of these, as we will see.

Definition 4.6

The Mobius function $\mu : \mathbb{N} \rightarrow \mathbb{Z}$ is given by:

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1 \\ 0 & \text{if } \exists p > 1 \text{ prime with } p^2 | n \\ (-1)^r & \text{if } n = p_1 \cdots p_r \text{ and } p_i \text{ are distinct primes} \end{cases}$$

Definition 4.7

We say that n is square free if there is no prime p such that $p^2 | n$.

Lemma 4.8

The Mobius function μ is multiplicative.

Proof

By unique factorization, we are allowed to assume that

$$n = ab \quad \text{where} \quad a = \prod_{i=1}^r p_i^{l_i} \quad \text{and} \quad b = \prod_{i=r+1}^s p_i^{l_i}$$

If a equals 1, then $\mu(ab) = \mu(a)\mu(b) = 1\mu(b)$, and similar if $b = 1$. If either a or b is not square free, then neither is $n = ab$, and so in that case, we again have $\mu(ab) = \mu(a)\mu(b) = 0$. If both a and b are square free, then r (in the definition of μ) is strictly additive and so $(-1)^r$ is strictly multiplicative, hence multiplicative.

Definition 4.9

Euler's phi function, also called Euler's totient function is defined as follows: $\phi(n)$ equals the number of integers in $\{1, \dots, n\}$ that are relative prime to n .

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4.2: Additive Functions

Also important are the additive functions to which we will return in Chapter ??.

Definition 4.10

An additive function is a sequence such that $\gcd(a, b) = 1$ implies $f(ab) = f(a) + f(b)$. A completely additive function is one where the condition that $\gcd(a, b) = 1$ is not needed.

Definition 4.11

Let $\omega(n)$ denote the number of distinct prime divisors of n and let $\Omega(n)$ denote the number of prime powers that are divisors of n . These functions are called the prime omega functions.

So if $n = \prod_{i=1}^s p_i^{l_i}$, then

$$\omega_n = s \quad \text{and} \quad \Omega(n) = \sum_{i=1}^s l_i$$

The additivity of ω and the complete additivity of Ω should be clear.

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4.3: Mobius inversion

Lemma 4.12

Define $\epsilon(n) \equiv \sum_{d|n} \mu(d)$. Then $\epsilon(1) = 1$ and for all $n > 1$, $\epsilon(n) = 0$.

Proof

Lemma 4.8 says that μ is multiplicative. Therefore, by Proposition 4.3, ϵ is also multiplicative. It follows that $\epsilon(\pi_{i=1}^r p_i^{l_i})$ can be calculated

by evaluating a product of terms like $\epsilon(p^l)$ where p is prime. For example, when p is prime, we have

$$\epsilon(p) = \mu(1) + \mu(p) = 1 + (-1) = 0$$

and

$$\epsilon(p^2) = \mu(1) + \mu(p) + \mu(p^2) = 1 - 1 + 0 = 0$$

Thus one sees that $\epsilon(p^l)$ is zero unless $l = 0$.

Lemma 4.13

For $n \in \mathbb{N}$, define

$$S_n \equiv \{(a, b) \in \mathbb{N}^2 \mid \exists d > 0 \text{ such that } d|n \text{ and } ab = d\}$$

$$T_n \equiv \{(a, b) \in \mathbb{N}^2 \mid b|n \text{ and } a|\frac{n}{b}\}$$

Then $S_n = T_n$.

Proof

Suppose (a, b) is in S_n . Then $ab|n$ and so

$$ab = d|n \Big\} \Rightarrow b|n \text{ and } a|\frac{n}{b}$$

And so (a, b) is in T_n . Vice versa, if (a, b) is in T_n , then by setting $d \equiv ab$,

we get

$$\left| a|\frac{n}{b} \right\} \Rightarrow d|n \text{ and } ab = d$$

And so (a, b) is in S_n

Theorem 4.14: Mobius Inversion

Let $F : \mathbb{N} \rightarrow \mathbb{C}$ be any number theoretic function. Then the equation

$$F(n) = \sum_{d|n} f(d)$$

if and only if $f : \mathbb{N} \rightarrow \mathbb{C}$ satisfies

$$f(d) = \sum_{a|d} \mu(a) F\left(\frac{d}{a}\right) = \sum_{\{(a,b)|ab=d\}} \mu(a) F(b)$$

Proof

$\Leftarrow$: We show that substituting f gives F . Define H as

$$H(n) \equiv \sum_{d|n} f(d) = \sum_{d|n} \sum_{a|d} \mu(a) F\left(\frac{d}{a}\right)$$

Then we need to prove that $H(n) = F(n)$. This proceeds in three steps. For the first step we write $ab = d$, so that now

$$H(n) \equiv \sum_{d|n} f(d) = \sum_{d|n} \sum_{ab=d} \mu(a) F(b)$$

For the second step we apply Lemma 4.13 to the set over which the summation takes place. This gives:

$$H(n) = \sum_{b|n} \sum_{a|(\frac{n}{b})} \mu(a) F(b) = \sum_{b|n} \left(\sum_{a|(\frac{n}{b})} \mu(a) \right) F(b)$$

Finally, Lemma 4.12 implies that the term in parentheses equals $G\left(\frac{n}{b}\right)$. It equals 0, except when $b = n$ when it equals 1.

The result follows.

Uniqueness: Suppose there are two solutions f and g . We have:

$$F(n) = \sum_{d|n} f(d) = \sum_{d|n} g(d)$$

We show by induction on n that $f(n) = g(n)$.

Clearly $F(1) = f(1) = g(1)$. Now suppose that for $i \in \{1, \dots, k\}$, we have $f(i) = g(i)$. Then

$$F(k+1) = \left(\sum_{d|(k+1), d \leq k} f(d) \right) + f(k+1) = \left(\sum_{d|(k+1), d \leq k} g(d) \right) + g(k+1)$$

So that the desired equality for $k+1$ follows from the induction hypothesis.

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4.4: Euler's Phi or Totient Function

Recall the phi function from Definition 4.9.

Lemma 4.15 (Gauss)

For $n \in \mathbb{N} : n = \sum_{d|n} \varphi(d)$.

Proof

Define $S(d, n)$ as the set of integers m between 1 and n such that $\gcd(m, n) = d$:

$$S(d, n) = \{m \in \mathbb{N} | m \leq n \text{ and } \gcd(m, n) = d\}$$

This is equivalent to

$$S(d, n) = \{m \in \mathbb{N} | m \leq n \text{ and } \gcd(md, dn) = 1\}$$

From the definition of Euler's phi function, we see that the cardinality $|S(d, n)|$ of $S(d, n)$ is given by $\varphi(\frac{n}{d})$. Thus we obtain:

$$n = \sum_{d|n} |S(d, n)| = \sum_{d|n} \varphi(\frac{n}{d})$$

As d runs through all divisors of n in the last sum, so does $\frac{n}{d}$. Therefore the last sum is equal to $\sum_{d|n} \varphi(d)$, which proves the lemma.

Theorem 4.16

Let $\prod_{i=1}^r p_i^{l_i}$ be the prime power factorization of n . Then

$$\varphi(n) = \prod_{i=1}^r (1 - \frac{1}{p_i})$$

Proof

Apply Mobius inversion to Lemma 4.15:

$$\varphi(n) = \sum_{d|n} \mu(d) \frac{n}{d} = n \sum_{d|n} \frac{\mu(d)}{d}$$

The functions μ and $d \rightarrow \frac{1}{d}$ are multiplicative. It is easy to see that the product of two multiplicative functions is also multiplicative. Therefore φ is also multiplicative (Proposition 4.3). Thus

$$\varphi(\prod_{i=1}^r p_i^{l_i}) = \prod_{i=1}^r \varphi(p_i^{l_i})$$

So it is sufficient to evaluate the function φ on prime powers. Noting that the divisors of the prime power p^l are $\{1, p, \dots, p^l\}$, we get from Equation

$$\varphi(p^l) = p^l \sum_{j=0}^l \frac{\mu(p^j)}{p^j} = p^l (1 - \frac{1}{p})$$

Substituting this into Equation 4.4 completes the proof.

From this proof we obtain the following corollary.

Corollary 4.17

Euler's phi function is multiplicative.

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4.5: Dirichlet and Lambert Series

We will take a quick look at some interesting series without worrying too much about their convergence, because we are ultimately interested in the analytic continuations that underlie these series. For that, it is sufficient that there is convergence in any open non-empty region of the complex plane.

Definition 4.18

Let f , g , and F be arithmetic functions (see Definition 4.1). Define the Dirichlet convolution of f and g , denoted by $f \star g$, as

$$(f \star g)(n) \equiv \sum_{ab=n} f(a)g(b)$$

This convolution is a very handy tool. Similar to the usual convolution of sequences, one can think of it as a sort of multiplication. It pays off to first define a few standard number theoretic functions.

Definition 4.19

We use the following notation for certain standard sequences. The sequence $\epsilon(n)$ is 1 if $n = 1$ and otherwise returns 0, $\mathbf{1}(n)$ always returns 1, and $I(n)$ returns n ($I(n) = n$).

The function ϵ acts as the identity of the convolution. Indeed, $(\epsilon \star f)(n) = \sum_{ab=n} \epsilon(a)f(b) = f(n)$.

Note that $I(n)$ is the identity as a function, but should not be confused with the identity of the convolution (ϵ). In other words, $I(n) = n$ but $I \star f \neq f$.

We can now do some very cool things of which we can unfortunately give but a few examples. As a first example, we reformulate the Möbius inversion of Theorem 4.14 as follows:

$$F = \mathbf{1} \star f \Leftrightarrow f = \mu \star F$$

This leads to the next example. The first of the following equalities holds by Lemma 4.15, the second follows from Möbius inversion.

$$I = \mathbf{1} \star \varphi \Leftrightarrow \varphi = \mu \star I$$

And the best of these examples is gotten by substituting the identity ϵ for F in equation (4.5):

$$\epsilon = \mathbf{1} \star f \Leftrightarrow f = \mu \star \epsilon = \mu$$

Thus μ is the convolution inverse of the sequence $(1, 1, 1, \dots)$. This immediately leads to an unexpected expression for $1/\zeta(s)$ of equation (4.8).

Definition 4.20

Let $f(n)$ is an arithmetic function (or sequence). A Dirichlet series is a series of the form $F(s) = \sum_{n=1}^{\infty} f(n)n^{-s}$. Similarly, a Lambert series is a series of the form $F(x) = \sum_{n=1}^{\infty} f(n) \frac{x^n}{1-x^n}$.

The prime example of a Dirichlet series is – of course – the Riemann zeta function of Definition 2.19, $\zeta(s) = \sum \mathbf{1}(n)n^{-s}$.

Theorem 4.21

For the product of two Dirichlet series we have

$$\left(\sum_{n=1}^{\infty} f(n)n^{-s} \right) \left(\sum_{n=1}^{\infty} g(n)n^{-s} \right) = \sum_{n=1}^{\infty} (f \star g)(n)n^{-s}$$

Proof

This follows easily from re-arranging the terms in the product:

$$\sum_{a,b \geq 1} \frac{f(a)g(b)}{(ab)^s} = \sum_{n=1}^{\infty} \left(\sum_{n=1}^{\infty} f(a)g(b) \right) n^{-s}$$

We collected the terms with $ab = n$.

Since $\frac{\zeta(s)}{\zeta(s)} = 1$, equation (4.7) and the lemma immediately imply that

$$\frac{1}{\zeta(s)} = \sum_{n \geq 1} \frac{\mu(n)}{n^s}$$

Recall from Chapter 2 that one of the chief concerns of number theory is the location of the non-real zeros of ζ . At stake is Conjecture 2.22 which states that all its non-real zeros are on the line $\text{Res} = 1/2$. The original definition of the zeta function is as a series that is absolutely convergent for $\text{Res} > 1$ only. Thus it is important to establish that the analytic continuation of ζ is valid for $\text{Res} \leq 1$. The next result serves as a first indication that $\zeta(s)$ can indeed be continued for values $\text{Res} \leq 1$.

Theorem 4.22

Let ζ be the Riemann zeta function and σ_k as in Definition 4.4, then

$$\zeta(s-k)\zeta(s) = \sum_{n=1}^{\infty} \frac{\sigma_k(n)}{n^s}$$

Proof

$$\zeta(s-k)\zeta(s) = \sum_{a \geq 1} a^{-s} \sum_{b \geq 1} b^k b^{-s} = \sum_{n \geq 1} n^{-s} \sum_{b|n} b^k$$

Lemma 4.23

A Lambert series can re-summed as follows:

$$\sum_{n=1}^{\infty} f(n) \frac{x^n}{1-x^n} = \sum_{n=1}^{\infty} (\mathbf{1} * f)(n) x^n$$

Proof

First use that

$$\frac{x^b}{1-x^b} = \sum_{a=1}^{\infty} x^{ab}$$

This gives that

$$\sum_{n=1}^{\infty} f(n) \frac{x^n}{1-x^n} = \sum_{b=1}^{\infty} \sum_{a=1}^{\infty} f(b) x^{ab}$$

Now set $n = ab$ and collect terms. Noting that $(\mathbf{1} * f)(b) = \sum_{b \mid n} f(b)$ yields the result.

Corollary 4.24

The following equality holds

$$\sum_{n \geq 1} \varphi(n) \frac{x^n}{1-x^n} = \frac{x}{(1-x)^2}$$

Proof

We have

$$\sum_{n \geq 1} \varphi(n) \frac{x^n}{1 - x^n} = \sum_{n \geq 1} (\mathbf{1} * \varphi)(n) x^n = \sum_{n \geq 1} I(n) x^n$$

The first equality follows from Lemma 4.23 and the second from Lemma 4.15. The last sum can be computed as $x \frac{d}{dx} (1 - x)^{-1}$ which gives the desired expression.

The last result is of importance in the study of dynamical systems. In figure 8, the map f_t is constructed by truncating the map $x \rightarrow 2x \bmod 1$ for $t \in [0, 1]$. Corollary 4.24 can be used to show that the set of t for which f_t does not have a periodic orbit has measure (“length”) zero [24, 25], even though that set is uncountable.

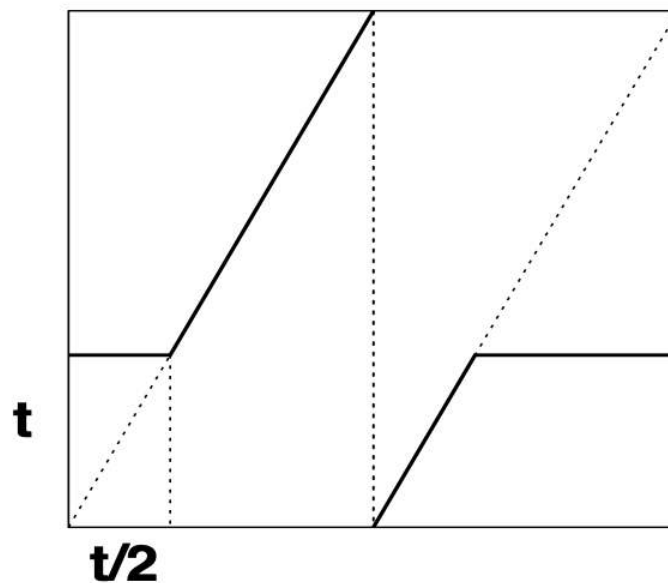


Figure 8. A one parameter family f_t of maps from the circle to itself. For every $t \in [0, 1]$ the map f_t is constructed by truncating the map $x \rightarrow 2x \bmod 1$ as indicated in this figure.

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4.6: Exercise

Exercise 4.6.1

Decide which functions are not multiplicative, multiplicative, or completely multiplicative (see Definition 4.2).

1. $f(n) = 1$.
2. $f(n) = 2$.
3. $f(n) = \sum_{i=1}^n i$.
4. $f(n) = \prod_{i=1}^n i$.
5. $f(n) = n$.
6. $f(n) = nk$.
7. $f(n) = \sum_{d|n} d$.
8. $f(n) = \prod_{d|n} d$.

Exercise 4.6.2

1. Let $h(n) = 0$ when n is even, and 1 when n is odd. Show that h is multiplicative.
2. Now let $H(n) = \sum_{d|n} h(d)$. Show without using Proposition 4.3 that H is multiplicative. (Hint: write $a = 2^k \prod_{i=1}^r p_i^{l_i}$ by unique factorization, where the p_i are odd primes. Similarly for b .)
3. What does Proposition 4.3 say?

Exercise 4.6.3

1. Compute the numbers $\sigma_1(n) = \sigma(n)$ of Definition 4.4 for $n \in \{1, \dots, 30\}$ without using Theorem 4.5.
2. What is the only value n for which $\sigma(n) = n$?
3. Show that $\sigma(p) = p + 1$ whenever p is prime.
4. Use (c) and multiplicativity of σ to check the list obtained in (a).
5. For what values of n in the list of (a) is $n | \sigma(n)$? (Hint: 6 and 28.)

Exercise 4.6.4

1. Compute the numbers $\sigma_0(n) = \tau(n)$ of Definition 4.4 for $n \in \{1, \dots, 30\}$ without using Theorem 4.5.
2. What is the only value n for which $\tau(n) = 1$?
3. Show that $\tau(p) = 2$ whenever p is prime.
4. Use (c) and multiplicativity of τ to check the list obtained in (a).

Exercise 4.6.5

1. Compute the numbers φ of Definition 4.9 for $n \in \{1, \dots, 30\}$ without using Theorem 4.16.
2. What is $\varphi(p)$ when p is a prime?
3. How many positive numbers less than pn are not divisible by p ?
4. Use (c) and multiplicativity of φ to check the list obtained in (a).

Exercise 4.6.6

1. Compute the numbers $\mu(n)$ of Definition 4.6 for $n \in \{1, \dots, 30\}$.
2. What is $\mu(p)$ when p is a prime?
3. Use (c) and multiplicativity of μ to check the list obtained in (a).

Exercise 4.6.7

Let $\tau(n)$ be the number of distinct positive divisors of n . Answer the following question without using Theorem 4.5.

1. Show that τ is multiplicative.
2. If p is prime, show that $\tau(p^k) = k + 1$.
3. Use the unique factorization theorem, to find an expression for $\tau(n)$ for $n \in \mathbb{N}$.

Exercise 4.6.8

Two positive integers a and b are called amicable if $\sigma(a) = \sigma(b) = a + b$. The smallest pair of amicable numbers is formed by 220 and 284.

1. Use Theorem 4.5 to show that 220 and 284 are amicable.
2. The same for 1184 and 1210.

Exercise 4.6.9

A positive integer n is called perfect if $\sigma(n) = 2n$.

1. Show that n is perfect if and only if the sum of its positive divisors less than n equals n .
2. Show that if p and $2^p - 1$ are primes, then $n = 2^{p-1}(2^p - 1)$ is perfect. (Hint: use Theorem 4.5 and exercise 4.3(c).)
3. Use exercise 1.14 to show that if $2^p - 1$ is prime, then p is prime, and thus $n = 2^{p-1}(2^p - 1)$ is perfect.
4. Check that this is consistent with the list in exercise 4.3.

Exercise 4.6.10

Draw the following directed graph G : the set of vertices V represent 0 and the natural numbers between 1 and 50. For $a, b \in V$, a directed edge ab exists if $\sigma(a) - a = b$. Finally, add a loop at the vertex representing 0. Notice that every vertex has 1 outgoing edge, but may have more than 1 incoming edge.

1. Find the cycles of length 1 (loops). The non-zero of these represent perfect numbers.
2. Find the cycles of length 2 (if any). A pair of numbers a and b that form a cycle of length 2 are called amicable numbers. Thus for such a pair, $\sigma(b) - b = a$ and $\sigma(a) - a = b$.
3. Find any longer cycles. Numbers represented by vertices in longer cycles are called sociable numbers.
4. Find numbers whose path ends in a cycle of length 1. These are called aspiring numbers.
5. Find numbers (if any) that have no incoming edge. These are called untouchable numbers.
6. Determine the paths starting at 2193 and at 562. (Hint: both end in a cycle (or loop).)

A path through this graph is called an aliquot sequence. The so-called Catalan-Dickson conjecture says that every aliquot sequence ends in some finite cycle (or loop). However, even for a relatively small number such as 276, it is unknown (in 2017) whether its aliquot sequence ends in a cycle.

Exercise 4.6.11

In this exercise, we give a different proof of Theorem 4.16. It uses the principle of inclusion-exclusion [21]. We state it here for completeness. Let S be a finite set with subsets A_1, A_2 , and so on through A_r . Then, if we denote the cardinality of a set A by $|A|$,

$$|S - \bigcup_{i=1}^r A_i| = |S| - |S_1| + |S_2| - \cdots + (-1)^r |S_r|$$

where $|S_l|$ is the sum of the sizes of all intersections of l members of $\{A_1, \dots, A_r\}$.

Now, in the following we keep to the following conventions. Using prime factorization, write

$$n = \prod_{i=1}^r p_i^{k_i}$$

$$A_i = \{z \in S \mid p_i \text{ divides } z\}$$

$$S = \{1, 2 \cdots n\} \text{ and } R = \{1, 2 \cdots r\}$$

$$I_l \subseteq R \text{ such that } |I_l| = l$$

1. Show that $\varphi(n) = |S - \bigcup_{i=1}^r A_i|$. (Hint: any number that is not co-prime with n is a multiple of at least one of the p_i .)
2. Show that $|A_i| = \frac{n}{p_i}$
3. Show that $|\bigcap_{i \in I_l} A_i| = n \prod_{i \in I_l} \frac{1}{p_i}$. (Hint: use Corollary 3.7.)
4. Show that $|S_l| = n \sum_{I_l \subseteq R} \prod_{i \in I_l} \frac{1}{p_i}$.
5. Show that the principle of inclusion-exclusion implies that $|S - \bigcup_{i=1}^r A_i| = n + n \sum_{l=1}^r (-1)^l \sum_{I_l \subseteq R} \prod_{i \in I_l} \frac{1}{p_i}$.
6. Show that $n + n \sum_{l=1}^r (-1)^l \sum_{I_l \subseteq R} \prod_{i \in I_l} \frac{1}{p_i} = n \prod_{i=1}^r (1 - \frac{1}{p_i})$. Notice that this implies Theorem 4.16. (Hint: write out the product $\prod_{i=1}^r (1 - \frac{1}{p_i})$.)

Exercise 4.6.12

Let $F(n) = n = \sum_{d|n} f(n)$. Use the Mobius inversion formula (or $f(n) = \sum_{d|n} \mu(d) F(\frac{n}{d})$) to find $f(n)$. (Hint: substitute the Mobius function of Definition 4.6 and use multiplicativity where needed.)

Exercise 4.6.13

1. Compute the sets S_n and T_n of Lemma 4.13 explicitly for $n = 4$ and $n = 12$.
2. Perform the resummation done in equations 4.1 and 4.2 explicitly for $n = 4$ and $n = 12$.

Exercise 4.6.14

Recall the definition of Dirichlet convolution $f * g$ of the arithmetic functions f and g . (Definition 4.18)

1. Show that Dirichlet convolution is commutative, that is

$$f * g = g * f$$

2. Show that Dirichlet convolution is associative, that is

$$(f * g) * h = g * (f * h)$$

3. Show that Dirichlet convolution is distributive, that is

$$(f * (g + h)) = f * g + f * h$$

4. The binary operation Dirichlet convolution has an identity ϵ , defined by

$$f * \epsilon = \epsilon * f = f$$

Show that the function ϵ of Lemma 4.12 is the identity of the convolution.

Exercise 4.6.15

Use exercise 4.14 to prove the following:

1. Show that the Dirichlet convolution of two multiplicative functions is multiplicative.
2. Show that the sum of two multiplicative functions is not necessarily multiplicative. (Hint: $\epsilon + \epsilon$.)

Exercise 4.6.16

See Definition 4.11. Define $f(n) \equiv \tau(n^2)$ and $g(n) \equiv 2^{\omega(n)}$

1. Compute $\omega(n)$, $f(n)$, and $g(n)$ for n equals 10^n and $6!$.
2. For p prime, show that $\tau(p^{2k}) = \sum_{d|p^k} 2^{\omega(d)} = 2k + 1$. (Hint: use Theorem 4.5.)

3. Show that f is multiplicative. (Hint: use that τ is multiplicative.)
4. Use (d) to show that g is multiplicative.
5. Show that

$$\tau(n^2) = \sum_{d|n} 2^{\omega(d)}$$

Exercise 4.6.17

Let $S(n)$ denote the number of square free divisors of n with $S(1) = 1$ and $\omega(n)$ the number of distinct prime divisors of n . See also Definition 4.11.

1. Show that $S(n) = \sum_{d|n} |\mu(d)|$. (Hint: use definition 4.6)
2. Show that $S(n) = 2^{\omega(n)}$. (Hint: let W be the set of prime divisors of n . Then every square free divisor corresponds to a subset-product- of those primes. How many subsets of primes are there in W ?)
3. Conclude that

$$\sum_{d|n} |\mu(d)| = 2^{\omega(n)}$$

Exercise 4.6.18

Define the Liouville λ -function by $\lambda(1) = 1$ and $\lambda(n) = (-1)^{\Omega(n)}$.

1. Compute $\lambda(10n)$ and $\lambda(6!)$.
2. Show that λ is multiplicative. (Hint: $\Omega(n)$ is completely additive.)
3. Use Proposition 4.3 to show that $F(n) = \sum_{d|n} \lambda(d)$ is multiplicative.
4. For p prime, show that

$$\sum_{d|p^k} \lambda(d) = \sum_{i=0}^k (-1)^i$$

which equals 1 if k is even and 0 if k is odd.

5. Use (c) and (d) to conclude that

$$F(n) = \sum_{d|n} \lambda(d) = \begin{cases} 1 & \text{if } n = m^2 \\ 0 & \text{else} \end{cases}$$

Exercise 4.6.19

Let f be a multiplicative function.

Define $q(n) \equiv \sum_{d|n} \mu(d)f(d)$, where μ is the Mobius function.

1. Show that $f(1) = 1$.
2. Show that $f\mu$ (their product) is multiplicative.
3. Use Proposition 4.3 to show that $q(n)$ is multiplicative.
4. Show that if p is prime, then $q(p^k) = f(1) - f(p) = 1 - f(p)$.
5. Use (c) and (d) to show that

$$q(n) = \sum_{d|n} \mu(d)f(d) = \prod_{p \text{ prime}, p|n} (1 - f(p))$$

Exercise 4.6.20

Use exercise 4.19 (e) and the definition of ω in exercise 4.16 and λ in exercise 4.18 to show that

$$\sum_{d|n} \mu(d)\lambda(d) = 2\omega(n)$$

Exercise 4.6.21

1. Show that for all $n \in \mathbb{N}$, $\mu(n)\mu(n+1)\mu(n+2)\mu(n+3) = 0$. (Hint: divisibility by 4.)
2. Show that for any integer $n \geq 3$, $\sum_{k=1}^n \mu(k!) = 1$. (Hint: use (a).)

Exercise 4.6.22

1. Use Euler's product formula and the sequence μ of Definition 4.6 to show that

$$\frac{1}{\zeta(s)} = \prod_{p \text{ prime}} (1 - p^{-s}) = \prod_{p \text{ prime}} \left(\sum_{i \geq 0} \mu(p^i) p^{-is} \right)$$

2. Without using equation (4.7), prove that the expression in (a) equals $\sum_{n \geq 1} \mu(n) n^{-s}$. (Hint: since μ is multiplicative, you can write a proof re-arranging terms as in the first proof of Euler's product formula.)

Exercise 4.6.23

1. Use equation (4.8) to show that

$$\zeta(s-1) = \sum_{a \geq 1} \frac{a}{a^s} \sum_{b \geq 1} \frac{\mu(b)}{b^s}$$

2. Use Lemma 4.23 and the first equality of equation (4.3) to show that

$$\frac{\zeta(s-1)}{\zeta(s)} = \sum_{n \geq 1} \varphi(n^s)$$

Exercise 4.6.24

1. Use Corollary 4.22 to show that

$$\zeta(s-k) = \sum_{a \geq 1} \frac{\sigma_k(a)}{a^s} \sum_{b \geq 1} \frac{\mu(b)}{b^s}$$

2. Show that

$$\zeta(s-k) = \sum_{n \geq 1} (\sigma_k * \mu)(n) n^{-s}$$

where $*$ means the Dirichlet convolution (Definition 4.18).

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CHAPTER OVERVIEW

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5.1: Modular Arithmetic

Modular Arithmetic concerns itself with computations involving addition and multiplication in $\mathbb{Z}$ module b , denoted by $\mathbb{Z}_b$, i.e., calculations with residues module b (See definition 1.6). One common way of looking at this is to consider integers x and y that differ by a multiple of b as equivalent (see exercise 5.1). We write $x \sim y$. One then proves that the usual addition and multiplication is well-defined for these equivalence classes. This is done in exercise 5.2.

Suppose we are given an operation $*$ on a set G and that $(G, *)$ satisfies certain requirements - namely that it forms a group - that include having an identity element e . The order of an element g is the smallest positive integer k such that $g * g \cdots g$, repeated k times and usually written g^k , equals e . One can show that the elements $\{e, g, g^2, g^{k-1}\}$ also form a group. To follow this up would take us too far a field. A few details are given in Chapter ???. The full story can be found in [10], [11], or [13].

In the case at hand, $\mathbb{Z}^b$, where we have a structure with two operations, namely addition with identity element 0 and multiplication with identity element 1. We could therefore define the order of an element in $\mathbb{Z}^b$ with respect to addition and with respect to multiplication. As an example, we consider the element 3 in $\mathbb{Z}^7$:

$$3 + 3 + 3 + 3 + 3 + 3 + 3 =_7 0 \quad \text{and} \quad 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 =_7 1$$

The first gives 7 as the additive order of 3, and the second gives 6 for the multiplicative order. For our current purposes, however, it is sufficient to work only with the multiplicative version.

Definition 5.1

The order of a modulo b , written as $\text{Ord}_b^\times(a)$, is the smallest positive number k such that $a^k \equiv_b 1$. (If there is no such k , the order is ∞ .)

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5.2: Euler's Method

Definition 5.2

A complete set of residues C modulo b is a set b integers in $\mathbb{Z}$, such C has exactly one integer in each congruence class (modulo b).

A reduced set of residues R modulo b is a set $\varphi(b)$ integers in $\mathbb{Z}$, such R has exactly one integer in each class congruent to $p \in \{1, \dots, b-1\}$ (modulo b) such that p is relatively prime to b .

Lemma 5.3

Suppose $\gcd(a, b) = 1$. If the numbers $\{x_i\}$ form a complete set of residues modulo b (a reduced set of residues modulo b), then $\{ax_i\}$ is a complete set of residues modulo b (a reduced set of residues modulo b).

Proof

Let $\{x_i\}$ be a complete set of residues modulo b . Then the b numbers $\{ax_i\}$ form complete set of residues unless two of them are congruent. But that is impossible by Theorem 2.7.

Let $\{x_i\}$ be a reduced set of residues modulo b . Then, as above, no two of the $\varphi(b)$ numbers $\{ax_i\}$ are congruent modulo b . Furthermore, Lemma 2.18 implies that if $\gcd(a, b) = 1$ and $\gcd(x_i, b) = 1$, then $\gcd(ax_i, b) = 1$. Thus the set $\{ax_i\}$ is a reduced set of residues modulo b .

Theorem 5.4 (Euler)

Let a and b greater than 0 and $\gcd(a, b) = 1$. Then $a^{\varphi(b)} \equiv 1$, where φ denotes Euler's phi or totient function (Definition 4.9).

Proof

Let $\{x_i\}_{i=1}^{\varphi(b)}$ be a reduced set of residues module b . Then by Lemma 5.3, $\{ax_i\}_{i=1}^{\varphi(b)}$ be a reduced set of residues module b . Because multiplication is commutative, we got

$$\prod_{i=1}^{\varphi(b)} x_i = b \prod_{i=1}^{\varphi(b)} ax_i = a^{\varphi(b)} \prod_{i=1}^{\varphi(b)} x_i$$

Since $\gcd(x_i, a) = 1$, Lemma 2.18 implies that $\gcd(\prod_{i=1}^{\varphi(b)} x_i, a) = 1$. The cancelation theorem applied to the equality between the first and third terms proves the result.

Euler's theorem says that $\varphi(b)$ is a multiple of $\text{Ord}_b^\times(a)$. But it does not say what multiple. In fact, in practice, that question is difficult to decide. It is of theoretical importance to decide when the two are equal.

Definition 5.5

Let a and b positive integers with $\gcd(a, b) = 1$. If $\text{Ord}_b^\times a = \varphi(b)$, then a is called a primitive root of b (or a primitive root module b).

For example, the smallest integer k for which $3^k \equiv 1 \pmod{7}$ is 6. Since $\varphi(7) = 6$, we see that 3 is primitive root of 7. Since multiplication is well-defined in $\mathbb{Z}_7$, it follows that $(3 + 7k)^6 \equiv 3^6 \equiv 1 \pmod{7}$. Thus, $\{\dots, -4, 3, 10, \dots\}$ are all primitive roots of 7. The only other non-congruent primitive root of 7 is 5. Not all numbers have primitive roots. For instance, 8 has none.

This has interesting connection with day-to-day arithmetic, namely the expression of rational numbers in base 10.

Proposition 5.6

Let a and n greater than 0 and $\gcd(a, n) = \gcd(10, n) = 1$. The decimal expansion of $\frac{a}{n}$ is non-terminating and eventually periodic with a period p , where (1) $p = \text{Ord}_n^\times(10)$ and (2) $p \mid \varphi(n)$.

Proof

The proof proceeds by excuting a long division, each step of which uses the division algorithm, Start by reducing a module n and call the result r_0 .

$$a = nq_0 + r_0$$

where $r_0 \in \{0, \dots, n-1\}$. Lemma 3.1 implies that $\gcd(a, n) = \gcd(r_0, n) = 1$. So in particular, $r_0 \neq 0$. The integer part of $\frac{a}{n}$ is q_0 . The next step of the long division is:

$$10r_0 = nq_1 + r_1$$

where again we choose $r_1 \in \{0, \dots, n-1\}$.

Note that $0 \leq 10r_0 < 10n$ and so $q_1 \in \{0, \dots, 9\}$. We now record the first digit “after the decimal point” of the decimal expansion: q_1 . By Lemma 3.1, we have $\gcd(10r_0, n) = \gcd(r_1, n)$. In turn, this implies via Lemma 2.18 that $\gcd(r_0, n) = \gcd(r_1, n)$. And again, we see that $r_1 \neq 0$.

The process now repeats itself.

$$\underbrace{10(10r_0 - nq_1)}_{r_1} = nq_2 + r_2$$

and we record the second digit after the decimal dot, $q_2 \in \{0, \dots, 9\}$. By the same reasoning, $\gcd(r_2, n) = 1$ and so $r_2 \neq 0$. One continues and proves by induction that $\gcd(r_i, n) = 1$. In particular, $r_i \neq 0$, so the expansion does not terminate.

Since the remainders r_i are in $\{1, \dots, n-1\}$, the sequence must be eventually periodic with (least positive) period p . At that point, we have

$$10^{k+p}r_0 =_n 10^k r_0$$

By Theorem 2.7, we can cancel the common factors 10^k and r_0 , and we obtain that $10^p =_n 1$. Since p is the least such (positive) number, we have proved part (1). Part (2) follows directly from Euler’s Theorem.

Of course, this proposition easily generalizes to computations in any other base b . As an en example, we mention that if $\gcd(a, n) = 1$ and b is a primitive root of n , then the expansion of $\frac{a}{b}$ has period $\varphi(n)$.

The next result is an immediate consequence of Euler’s theorem. It follows by setting $y = x + k\varphi(b)$. It has important applications in cryptography.

Corollary 5.7

Let a and b be coprime with $b \geq 0$. If $x =_{\varphi(b)} y$, then $a^x =_b a^y$.

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5.3: Fermat's Little Theorem and Primality Testing

Euler's theorem has many other important consequences. It implies what is known as Fermat's little theorem, although it was not proved by Fermat himself, since, as he writes in the letter in which he stated the result, he feared "going on too long". Not an isolated case, it would appear!

Corollary 5.8 (Fermat's Little Theorem)

If p is prime and $\gcd(a, p) = 1$, then $a^{p-1} \equiv_p 1$.

This follows from Euler's Theorem by noticing that for a prime p , $\varphi(p) = p - 1$. There is an equivalent formulation which allows p to be a divisor of a . Namely, if p is prime, then $a^p \equiv_p a$. Notice that if $p|a$, then both sides are congruent to 0.

Primes are of great theoretical and practical value (think of encryption, for example). Algorithms for primality testing are therefore very useful. The simplest test to find out if some large number $\sqrt{n}$ is prime, consists of course of applying some version of Eratosthenes' sieve to the positive integers less than or equal to n . By the prime number theorem (Theorem 2.21), this will give on the order of $\frac{2\sqrt{n}}{\ln n}$ primes. Then divide n by all these numbers. This will take on the order of $\frac{2\sqrt{n}}{\ln n}$ divisions.

Another possibility is to use the converse of Fermat's little theorem (Corollary 5.8). If n and p are distinct primes, we know that $p^{n-1} \equiv_n 1$. The Fermat primality test for n consists of testing for example whether $2^{n-1} \equiv_n 1$. However, the converse of Fermat's little theorem is not true! So even if $2^{n-1} \equiv_n 1$, it could be that n is not prime; we will discuss this possibility at the end of this section. As it turns out, primality testing via Fermat's little theorem can be done much faster than the naive method, provided one uses fast modular exponentiation algorithms. We briefly illustrate this technique by computing 11^{340} modulo 341.

Start by expanding 340 in base 2 as done in exercise 3.18, where it was shown that this takes on the order of $\log_2 340$ (long) divisions.

$$340 = 170 \cdot 2 + 0$$

$$170 = 85 \cdot 2 + 0$$

$$85 = 42 \cdot 2 + 1$$

$$42 = 21 \cdot 2 + 0$$

$$21 = 10 \cdot 2 + 1$$

$$10 = 5 \cdot 2 + 0$$

$$5 = 2 \cdot 2 + 1$$

$$2 = 1 \cdot 2 + 0$$

$$1 = 0 \cdot 2 + 1$$

And so

$$340 = 101010100 \text{ in base 2}$$

Next, compute a table of powers 11^{2^i} modulo 341, as done below. By doing this from the bottom up, this can be done using very few computations. For instance, once $11^8 \equiv_{341} 143$ has been established, the next up is found by computing 143^2 modulo 341, which gives 330, and so on. So this takes about $\log_2 340$ multiplications.

$$\begin{array}{rcl}
 1 & 11^2 56 & =_{341} 330 \\
 0 & 11^1 28 & =_{341} 143 \\
 1 & 11^{64} & =_{341} 319 \\
 0 & 11^{32} & =_{341} 121 \\
 1 & 11^{16} & =_{341} 330 \\
 0 & 11^8 & =_{341} 143 \\
 1 & 11^4 & =_{341} 319 \\
 0 & 11^2 & =_{341} 121 \\
 0 & 11^1 & =_{341} 11
 \end{array} \tag{5.3.1}$$

The first column in the table thus obtained now tells us which coefficients in the second we need to compute the result.

$$11^{340} =_{341} 330 \cdot 319 \cdot 330 \cdot 319 =_{341} 132$$

Again, this takes no more than $\log_2 340$ multiplications. Thus altogether, for a number x and a computation in base b , this takes on the order of $2\log_b x$ multiplications plus $\log_b x$ divisions. For large numbers, this is much more efficient than the $\sqrt{x}$ of the naive method.

The drawback is that we can get false positives.

Definition 5.9

The number $n \in \mathbb{N}$ is called a pseudoprime to the base a if $\gcd(a, n) = 1$ and $a^{n-1} =_n 1$ but nonetheless n is composite. (When the base is 2, the clause to the base 2 is often dropped.)

Some numbers pass all tests to every base and are still composite. These are called Carmichael numbers. The smallest Carmichael number is 561. It has been proved [7] that there are infinitely many of them.

Definition 5.10

The number $n \in \mathbb{N}$ is called a Carmichael number if it is composite and it is a pseudoprime to every base.

The smallest pseudoprime is 341, because $2^{340} =_{341} 1$ while $341 = 11 \cdot 31$. In this case, one can still show that 341 is not a prime by using a different base $3^{340} =_{341} 56$. By Fermat's little theorem, 341 cannot be prime.

The reason that the method sketched here is still useful is that pseudoprime are very much rarer than primes. The number below $2.5 \cdot 10^{10}$ contain on the order of 10^9 primes. At the same time, this set contains only 21853 pseudoprimes to the prime 2. There are only 1770 integers below $2.5 \cdot 10^{10}$ that are pseudoprime to the base 2, 3, 5, and 7. Thus if a number passes these four tests, it is overwhelmingly likely that it is a prime.

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5.4: Fermat and Mersenne Primes

Through the ages, back to early antiquity, people have been fascinated by numbers, such as 6, that are the sum of their divisors other than itself, to wit: $6 = 1 + 2 + 3$. Mersenne and Fermat primes, primes of the form $2^k \pm 1$, have attracted centuries of attention. Note that if p is a prime other than 2, then $p^k \pm 1$ is divisible by 2 and therefore not a prime.

Definition 5.11

1. The Mersenne primes are $M_k = 2^k - 1$. A Mersenne prime is a Mersenne number that is also prime.
2. The Fermat primes are $F_k = 2^{2^k} - 1$. A Fermat prime is a Fermat number that is also prime. The number $n \in \mathbb{N}$ is called perfect, if $\sigma(n) = 2n$

Lemma 5.12

1. If $ad = k$, then $2^d - 1 \mid 2^k - 1$.
2. If $ad = k$ and a is odd, then $2^d + 1 \mid 2^k + 1$.

Proof

We only prove (2). The other statement can be proved similarly. So suppose that a is odd, then

$$2^d = 2^{d+1} - 1 \Rightarrow 2^{ad} = 2^{d+1} (-1)^a = 2^{d+1} - 1 \Rightarrow 2^{ad} + 1 = 2^{d+1} - 1$$

which proves the statement. Notice that this includes the case where $d = 1$. In that case, we have $3 \mid 2^a + 1$ (whenever a odd).

A proof using the geometric series can be found in exercise 1.14. This lemma immediately implies the following.

Corollary 5.13

- If $2^k - 1$ is prime, then k is prime.
 If $2^k + 1$ is prime, then $k = 2^r$.

So, candidates of Mersenne primes are the numbers $2^p - 1$ where p is prime. This works for $p \in \{0, 1, \dots, 10\}$, but $2^{11} - 1$ is the monkey-wrench.

It is equal to $23 \cdot 89$ and thus is composite. After that, the Mersenne primes become increasingly sparse. Among the first approximately 2.3 million primes, only 45 give Mersenne primes. As of this writing (2017), it is not known whether there are infinitely many Mersenne primes. In 2016, a very large Mersenne prime was discovered: $2^{74,207,281} - 1$. Mersenne primes are used in pseudo-random number generators.

Turing to primes of the form $2^k + 1$, the only candidates are $F_r = 2^{2^r} + 1$. Fermat himself noted that F_r is prime for $0 \leq r \leq 4$, and he conjectured that all these numbers were primes. Again, Fermat did not quite get it right! It turns out that the 5-th Fermat number, $2^{32} + 1$, is divisible by 641 (see exercise 5.11). In fact, as of this writing in 2017, there are no other known Fermat primes among the first 297 297 Fermat numbers! Fermat primes are also used in pseudorandom number generators.

Lemma 5.14

If $2^k - 1$ is prime, then $k > 1$ and $2^{k-1}(2^k - 1)$ is perfect.

Proof

If $2^k - 1$ is prime, then it must be at least 2, and so $k > 1$. Let $n = 2^{k-1}(2^k - 1)$. Since σ is multiplicative and $2^k - 1$ is prime, we can compute (using Theorem 4.5):

$$\sigma(n) = \sigma(2^{k-1})\sigma(2^k - 1) = \left(\sum_{i=0}^{k-1} 2^i\right)2^k = (2^k - 1)2^k = 2n$$

which proves the lemma.

Theorem 5.15 (Euler's Theorem)

Suppose n is even. Then n is perfect if and only if n is of the form $2^{k-1}(2^k - 1)$ where $2^k - 1$ is prime.

Proof

One direction follows from the previous lemma. We only need to prove that if an even number $n = 2^{k-1}$ where $k \geq 2$, is perfect then it is of the form stipulated.

Since n is even, we may assume $n = 2^k$ where $k \geq 2$ and $2 \nmid q$. Using multiplicativity of σ and the fact that $\sigma(n) = 2n$:

$$\sigma(n) = \sigma(2^k)(2^k - 1) = 2^k$$

Thus

$$(2^k - 1)\sigma(2^k) - 2^k = 0$$

Since $2^k - (2^k - 1) = 1$, we know by Bezout that $\gcd((2^k - 1), 2^k) = 1$. Thus Proposition 3.5 implies that the general solution of the above equation is:

$$q = (2^k - 1)t \quad \text{and} \quad \sigma(q) = 2^k t$$

where $t > 0$, because we know that $q > 0$

Assume first that $t > 1$. The form of q , namely $q = (2^k - 1)t$, allow us to identify at least four distinct divisors of q . This gives that

$$\sigma(q) \geq 1 + t + (2^k - 1) + (2^k - 1)t = 2^k(t + 1)$$

This contradicts equation 5.2, and so $t = 1$.

Now use equation 5.2 again (with $t = 1$) to get that $n = 2^{k-1}(2^k - 1)$ has the required form. Furthermore, the same equation says that $\sigma(q) = \sigma(2^k - 1) = 2^k$ which proves that $2^k - 1$ is prime.

It is unknown at the date of this writing whether any odd perfect numbers exist.

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5.5: Division in $\mathbb{Z}_b$

The next result is a game changer! It tells us that there is a unique element a^{-1} such that $aa^{-1} =_b 1$ if and only if a is in the reduced set of residues (modulo b). Thus division is well-defined in the reduced set of residues modulo b . This serves as a clear signal to emphasize two distinct algebraic structures. A ring is a structure with addition and its inverse subtraction plus multiplication, but where multiplication may not have an inverse. A field is almost the same, but now multiplication always has an inverse: division. A more detailed description of these algebraic construction is given in Section ???. The numbers 1 and -1 are always in the reduced set of residues modulo b . This set is sometimes called the set of units (see Definition ???) of $\mathbb{Z}_b$.

Proposition 5.16

Let $\mathbb{R}$ be a reduced set of residues modulo b . Then

1. for every $a \in \mathbb{R}$, there is a unique a' in $\mathbb{R}$ such that $a'a =_b 1$
2. for every $a \notin \mathbb{R}$, there exists no $x \in \mathbb{Z}_b$ such that $ax =_b 1$
3. let $\mathbb{R} = \{x_i\}_{i=1}^{\varphi(b)}$, then also $\mathbb{R} = \{x-1\}_{i=1}^{\varphi(b)}$.

Proof

Statement 1: The existence of a solution follows immediately from Be'zout's Lemma, namely $a' =_b x$ solves for x in $ax + by = 1$. This solution must be in $\mathbb{R}$, because a , in turn, is the solution of $a'x + by = 1$ and thus Be'zout's Lemma implies that $\gcd(a', b) = 1$. Suppose we have two solutions $ax =_b 1$ and $ay =_b 1$, then uniqueness follows from applying the cancellation Theorem 2.7 to the difference of these equations.

Statement 2: By hypothesis, $\gcd(a, b) > 1$. We have that $ax =_b 1$ is equivalent to $ax + by = 1$, which contradicts Be'zout's Lemma.

Statement 3: This is similar to Lemma 5.3. By (1), we know that all inverses are in $\mathbb{R}$. So if the statement is false, there must be two elements of $\mathbb{R}$. So if the statement is false, there must be two elements of $\mathbb{R}$ with the same inverse: $ax =_b cx$. This is impossible by cancellation.

Lemma 5.17

Let p be prime. Then $a^2 =_p 1$ if and only if $a =_p \pm 1$

Proof

We have

$$a^2 =_p 1 \Leftrightarrow a^2 - 1 =_p (a+1)(a-1) =_p 0 \Leftrightarrow p|(a+1)(a-1)$$

Because p is prime, Corollary 2.12 says that either $p|a+1$ (and so $a =_p -1$) or $p|a-1$ (and so $a =_p +1$).

Perhaps, surprisingly, this last lemma is false if p is not prime. For example, $4^2 =_{15} 1$, but $4 \neq_{15} \pm 1$.

Theorem 5.18 (Wilson's theorem)

If p prime in $\mathbb{Z}$, then $(p-1)! =_p -1$. If b is composite, then $(b-1)! \neq_b -1$

Proof

This is true for $p = 2$. If $p > 2$, then Proposition 5.16(3) and Lemma 5.17 imply that every factor a_i in the product $(p-1)!$ other than -1 or 1 has a unique inverse a'_i different from itself. The factors a'_i run through all factors 2 through $p-2$ exactly once. Thus in the product, we can pair each a_i different from ± 1 with its inverse. This gives

$$(p-1)! =_p (+1)(-1) \prod a_i a'_i =_p -1$$

The second part is easier. If b is composite, there are least residues a and b greater than 1 so that $ad =_b 0$. Now either we can choose a and b distinct and then $(b-1)!$ contains the product ad , and thus it equals zero mod b . Or else this is

impossible and $b = a^2$. But then still $\gcd((b-1)!, b) \geq a$. By Bézout, we must have $(b-1)! \bmod b$ must be a multiple of a .

Wilson's theorem could be used to test primality of a number n . However, this takes n multiplications, which in practice is more expensive than trying to divide n by all numbers less than $\sqrt{n}$. Note, however, that if you want to compute a list of all primes between 1 and N , Wilson's theorem can be used much more efficiently. After computing $(k-1)! \bmod k$ to determine whether k is prime, it takes only 1 multiplication and 1 division to determine whether $k+1$ is prime.

Here is the take-away that will be important for Chapter ?? . More in particular, we have the following result.

Corollary 5.19

Let p be prime.

For every $a \in \mathbb{Z}_p$ there is a unique $a' \equiv_p -a$ such that $a + a' \equiv_p 0$.

For every $a \in \mathbb{Z}_p$ and $a \neq 0$, there is a unique $a' = a^{-1}$ so that $aa' \equiv_p 1$.

Addition and multiplication are well-defined in $\mathbb{Z}_b$ (see exercises 5.1 and 5.2). Thus when p is prime, we can add, multiply, subtract, and divide in $\mathbb{Z}_p$. In the words of Chapter ??, when p is a prime, then $\mathbb{Z}_p$ is a field. It is an interesting fact that the same is not true for a composite number b . According to Proposition 5.16, we need the reduced set of residues for the multiplication to be invertible. At the same time, for the set to be closed under multiplication, we need all of $\mathbb{Z}_b$ (think of $1+1+\dots$). Thus the operations addition and multiplication in $\mathbb{Z}_b$ collaborate with each other only if b is a prime.

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5.6: Exercises

Exercise 5.6.1

1. Let $m > 0$. Show that $a \equiv_m b$ is an equivalence relation on $\mathbb{Z}$. (Use Definitions 1.5 and 1.24.)
2. Describe the equivalence classes of $\mathbb{Z}$ modulo 6. (Which numbers in $\mathbb{Z}$ are equivalent to 0? Which are equivalent to 1? Et cetera.)
3. Show that the equivalence classes are identified by their residue, that is: $a \sim b$ if and only if $\text{Res}_m(a) = \text{Res}_m(b)$.

Note: If we pick one element of each equivalence class, such an element is called a representative of that class. The smallest non-negative representative of a residue class in $\mathbb{Z}_m$, is called the least residue. The collection consisting of the smallest non-negative representative of each residue class is called a complete set of least residues.

Exercise 5.6.2

This exercise relies on exercise 5.1. Denote the set of equivalence classes of $\mathbb{Z}$ modulo m by $\mathbb{Z}_m$ (see Definition 1.5). Prove that addition and multiplication are well-defined in $\mathbb{Z}_m$, using the following steps.

1. If $a \equiv_m a'$ and $b \equiv_m b'$, then $\text{Res}_m(a) + \text{Res}_m(b) \equiv_m \text{Res}_m(a') + \text{Res}_m(b')$. (Hint: show that $a + b = c$ if and only if $a + b \equiv_m c$. In other words: the sum modulo m only depend on $\text{Res}_m(a)$ and $\text{Res}_m(b)$ and not on which representative in the class (see exercise 5.1) you started with.)
2. Do the same for multiplication.

Exercise 5.6.3

Let $n = \sum_{i=1}^k a_i 10^i$ where $a_i \in \{0, 1, 2, \dots, 9\}$.

1. Show that $10^k \equiv_3 1$ for all $k \geq 0$. (Hint: use exercise 5.2.)
2. Show that $n \equiv_3 \sum_{i=1}^k a_i$.
3. Show that this implies that n is divisible by 3 if and only the sum of its digits is divisible by 3.

Exercise 5.6.4

Let $n = \sum_{i=1}^k a_i 10^i$ where $a_i \in \{0, 1, 2, \dots, 9\}$. Follow the strategy in exercise 5.3 to prove the following facts.

1. Show that n is divisible by 5 if and only if a_0 is. (Hint: Show that $n \equiv_5 a_0$.)
2. Show that n is divisible by 2 if and only if a_0 is.
3. Show that n is divisible by 9 if and only if $\sum_{i=1}^k a_i$ is.
4. Show that n is divisible by 11 if and only if $\sum_{i=1}^k (-1)^i a_i$ is.
5. Find the criterion for divisibility by 4.
6. Find the criterion for divisibility by 7. (Hint: this is a more complicated criterion!)

Exercise 5.6.5

1. List $(n-1)! \mod n$ for $n \in \{2, \dots, 16\}$.
2. Where does the proof of the first part of Wilson's theorem fail in the case of $n = 16$?
3. Does Wilson's theorem hold for $p = 2$? Explain!

Exercise 5.6.6

Denote by $\{x_i\}$ the reduced set of residues modulo 16. If the reasoning similar to that of Wilson's theorem holds, you expect that $\prod_i x_i \equiv_{16} -1$.

1. Compute $\prod_i x_i$.
2. How does the reasoning in Wilson's theorem fail? (Hint: see lemma 5.17.)
3. List the product of the reduced set of residues modulo n for $n \in \{2, \dots, 16\}$.

Exercise 5.6.7

1. For i in $\{1, 2, \dots, 11\}$ and j in $\{2, 3, \dots, 11\}$ make a table of $\text{Ord}_j^\times(i)$, i varying horizontally. After the j th column, write $\varphi(j)$.
2. List the primitive roots i modulo j for i and j as in (a). (Hint: the smallest primitive roots modulo j are: $\{1, 2, 3, 2, 5, 3, \emptyset, 2, 3, 2\}$)

Exercise 5.6.8

1. Determine the period of the decimal expansion of the following numbers: $100/13$, $13/77$ and $1/17$ through long division.
2. Use Proposition 5.6 to determine the period.
3. Check that this period equals a divisor of $\varphi(n)$.
4. The same questions for expansions in base 2 instead of base 10.

Exercise 5.6.9

1. Compute $2^{n-1} \bmod n$ for n odd in $\{3 \dots 40\}$.
2. Are there any pseudo-primes in the list?

Exercise 5.6.10

Assume that n is a pseudoprime to the base 2.

1. Show that $2^n - 2 =_n 0$.
2. Show from (a) that $n \mid M_n - 1$. (See Definition 5.11.)
3. Use Lemma 5.12 to show that (b) implies that $M_n \mid 2^{M_n-1} - 1$.
4. Conclude from (c) that if n is a pseudoprime in base 2, so is M_n .

Exercise 5.6.11

We show that the 5-th Fermat number, $2^{32} + 1$, is a composite number.

1. Show that $2^4 =_{641} -5^4$. (Hint: add 2^4 and 5^4 .)
2. Show that $2^{75} =_{641} -1$.
3. Show that $2^{32} + 1 = (2^7)^4 2^4 + 1 =_{641} 0$.
4. Conclude that F_5 is divisible by 641.

Exercise 5.6.12

1. Compute $\varphi(100)$. (Hint: use Theorem 4.16.)
2. Show that $179^{121} =_{100} 79^{121}$.
3. Show that $79^{121} =_{100} 79^1$. (Hint: use Theorem 5.4)
4. What are the last 2 digits of 179^{121} ?

The following 5 exercises on basic cryptography are based on [22]. First some language. The original readable message is called the plain text. Encoding the message is called encryption. And the encoded message is often called the encrypted message or code. To revert the process, that is: to turn the encrypted message back into plain text, you often need a key. Below we will encode the letters by 0 through 25 (in alphabetical order). We encrypt by using a multiplicative cipher. This means that we will encrypt our text by multiplying each number by the cipher modulo 26, and then return the corresponding letter. For example, if we use the cipher 3 to encrypt the plain text bob, we obtain the encrypted text as follows $1.14.1 \rightarrow 3.42.3 \rightarrow 3.16.3$.

Exercise 5.6.13

1. Use the multiplicative cipher 3 to decode DHIM.
2. Show that an easy way to decode is multiplying by 9 (modulo 26). The corresponding algorithm at the number level is called division by 3 modulo 26.
3. Suppose instead that our multiplicative cipher was 4. Encode bob again.
4. Can we invert *this* encryption by using multiplication modulo 26? Explain why.

Exercise 5.6.14

Suppose we have an alphabet of q letters and we encrypt using the multiplicative cipher $p \in \{0, \dots, q-1\}$. Use modular arithmetic to show that the encryption can be inverted if and only $\gcd(p, q) = 1$. (Hint: Assume the encryption of j_1 and j_2 are equal. Then look up and use the Unique Factorization theorem in Chapter 2.)

Exercise 5.6.15

Assume the setting of exercise 5.14. Assume p and q are such that the encryption is invertible. What is the decryption algorithm? Prove it. (Hint Find $r \in \{0, \dots, q-1\}$ such that $rp \equiv_q 1$. Then multiply the encryption by r .)

Exercise 5.6.16

Work out the last two problems if we encrypt using an affine cipher (a, p) . That is, the encryption on the alphabet $\{0, \dots, q-1\}$ is done as follows:

$$i \rightarrow a + pi \bmod q$$

Work out when this can be inverted, and what the algorithm for the inverse is.

Exercise 5.6.17

Decrypt the code V'ir Tbg n Frperg.

Exercise 5.6.18

1. Compute $7^{72} \bmod 13$, using modular exponentiation.
2. Similarly for $484^{187} \bmod 1189$.
3. Find $100! + 102! \bmod 101$. (Hint: Fermat.)
4. Show that $1381! \equiv_{1382} 0$. (Hint: Wilson.)

Exercise 5.6.19

Characterize the set of $n \geq 2$ for which $(n-1)! \bmod n$ is not in $\{0, -1\}$. (Hint: Wilson.)

Theorem 5.20 (Binomial Theorem)

If n is a positive integer, then

$$(a+b)^n = \sum_{i=0}^n \binom{n}{i} a^i b^{n-i} \quad \text{where} \quad \binom{n}{i} = \frac{n!}{i!(n-i)!}$$

Exercise 5.6.20

1. If p is prime, show that $\binom{p}{i} \bmod p$ equals 0 if $1 \leq i \leq p-1$, and equals 1 if $i = 0$ or $i = p$.
2. Evaluate $\binom{4}{2} \bmod 4$ and $\binom{6}{4} \bmod 6$. So, where in (a) did you use the fact that p is prime?
3. Use the binomial theorem to show that if p is prime $(a+b)^p \equiv_p a^p + b^p$

Exercise 5.6.21

Let p be prime.

1. Show that $0^p =_p 0$
2. Show that for $k > 0$, if $k^p =_p k$, then $(k+1)^p =_p k+1$. (Hint: use exercise 5.20.)
3. Conclude that for all $n \in \{0, 1, 2, \dots\}$, $n^p =_p n$. (Hint: use induction)
4. Prove that for all $n \in \mathbb{Z}$, $n^p =_p n$. (Hint: $(-n)^p =_p (-1)^p n^p$.)
5. Use (d) to prove Fermat's Little Theorem. (Hint: use cancellation.)

Exercises 3.19 and 3.22 show how to solve linear congruences generally. Quadratic congruences are much more complicated. As an example, we look at the equation $x^2 =_p \pm 1$ in the following exercise.

Exercise 5.6.22

1. Show that Fermat's little theorem gives a solution of $x^2 - 1 =_p 0$ whenever p is an odd prime. (Hint: consider $a^{\frac{p-1}{2}}$.)
2. Show that Wilson's theorem implies that for odd primes p

$$(-1)^{\frac{p-1}{2}} \left(\left(\frac{p-1}{2} \right)! \right)^2 =_p -1$$

(hint: the right hand gives all reduced residues module p .)

3. Show that if also $p \equiv_4 1$ (examples are 13, 17, 29, etc), then

$$\left(\left(\frac{p-1}{2} \right)! \right)^2 =_p -1$$

satisfies the quadratic congruence $x^2 + 1 =_p 0$.

4. Suppose $x^2 + 1 =_p 0$ and p is an odd prime. Show that Fermat's little theorem implies that

$$(-1)^{\frac{p-1}{2}} = 1$$

5. Show that p in (d) cannot satisfy $p \equiv_4 3$.

Exercise 5.6.23

Let $R = \{1, 3, 5, 7, 9, 11, 13, 15\}$ the reduced set of residues module 16. In $\mathbb{Z}_{16}$ and for each $r \in R$, define multiplication by r^{-1} (or division by r).

Exercise 5.6.24

1. Draw up the addition and multiplication tables for $\mathbb{Z}_7$.
2. Do the same for $\mathbb{Z}_8$. Is division well-defined.
3. The same for the reduced set of residues plus 0 in $\mathbb{Z}_8$. What goes wrong here?

Exercise 5.6.25

Given $n > 2$, let $R \subseteq \mathbb{Z}_n$ be the reduced set of residues and let $S \subseteq \mathbb{Z}_n$ be the set of solutions in $\mathbb{Z}_n$ of $x^2 =_n 1$ (or self inverses).

1. Show that $S \subseteq R$. (Hint: Be'zout)
2. Show that

$$\prod_{x \in R} x =_n \prod_{x \in S} x (= 1 \text{ if } S \text{ is empty})$$

3. Show that if S contains a , then it contains $-a$.
4. Show that if $a =_n -a$, then a and $-a$ are not in S .
5. Show that

$$\prod_{x \in R} x =_n (-1)^m \quad \text{some } m$$

6. Show that $m = |S|/2$.

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6.1: The Gauss Map

Definition 6.1

The Gauss map is the transformation $T : [0, 1] \rightarrow [0, 1)$ defined by

$$T(\xi) = \frac{1}{\xi} - \lfloor \frac{1}{\xi} \rfloor = \{ \frac{1}{\xi} \} \quad \text{and} \quad T(0) = 0$$

where we have used the notation of Definition 2.1.

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